

Universidade do Minho
Escola de Engenharia

Redes Fixas e Móveis (Fixed and Mobile Networks) 2026/2027

CCN
COMPUTER COMMUNICATIONS AND NETWORKS

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The Internet: a "nuts and bolts" view



Billions of connected computing *devices*:

- *hosts = end systems*
- running *network apps* at Internet's "edge"



Packet switches: forward packets (chunks of data)

- *routers, switches*

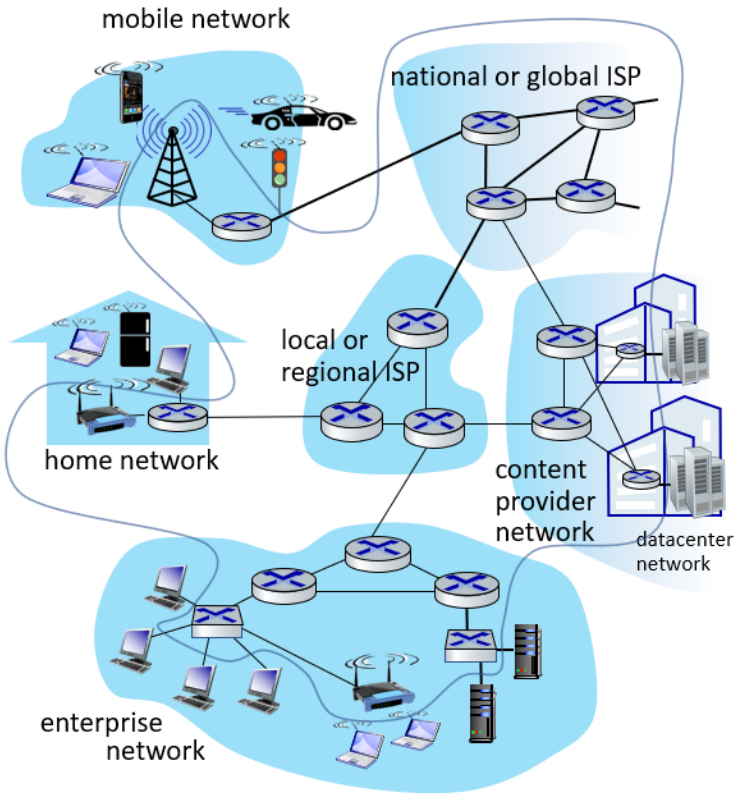


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

Networks

- collection of devices, routers, links: managed by an organization



Slide from Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross

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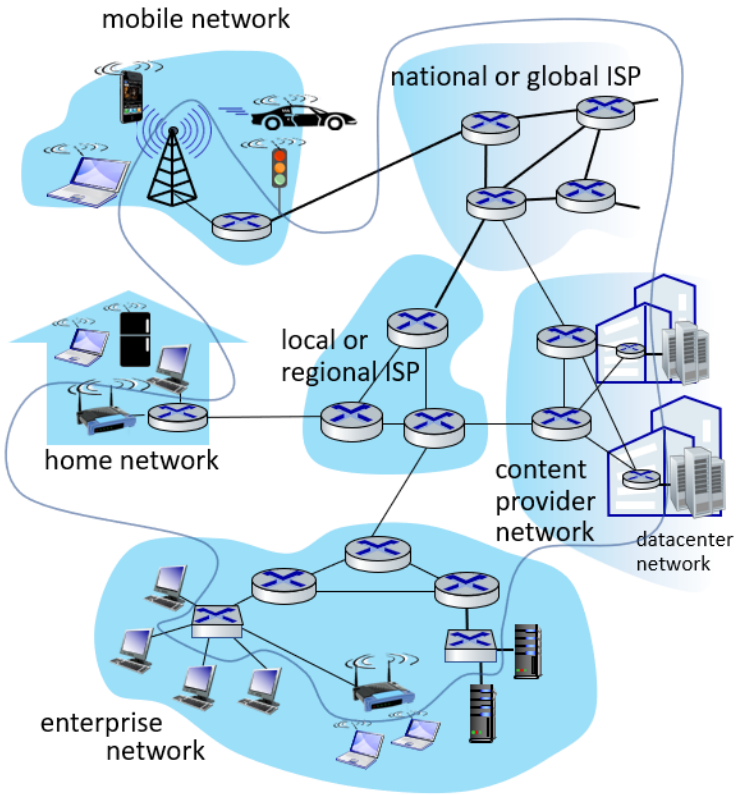


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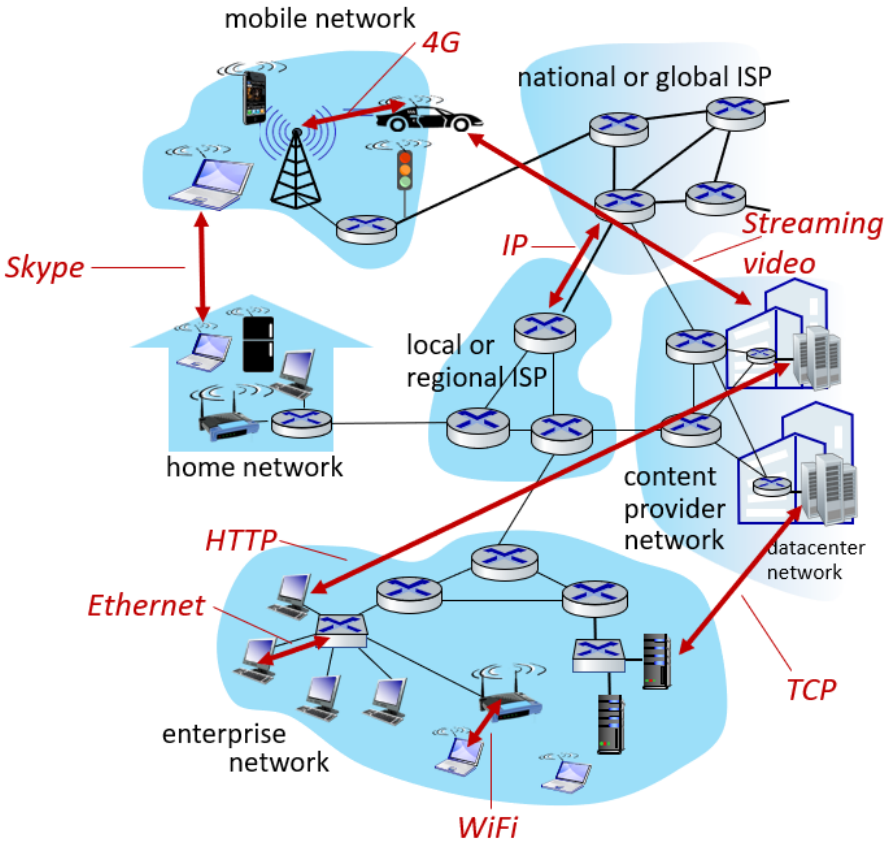
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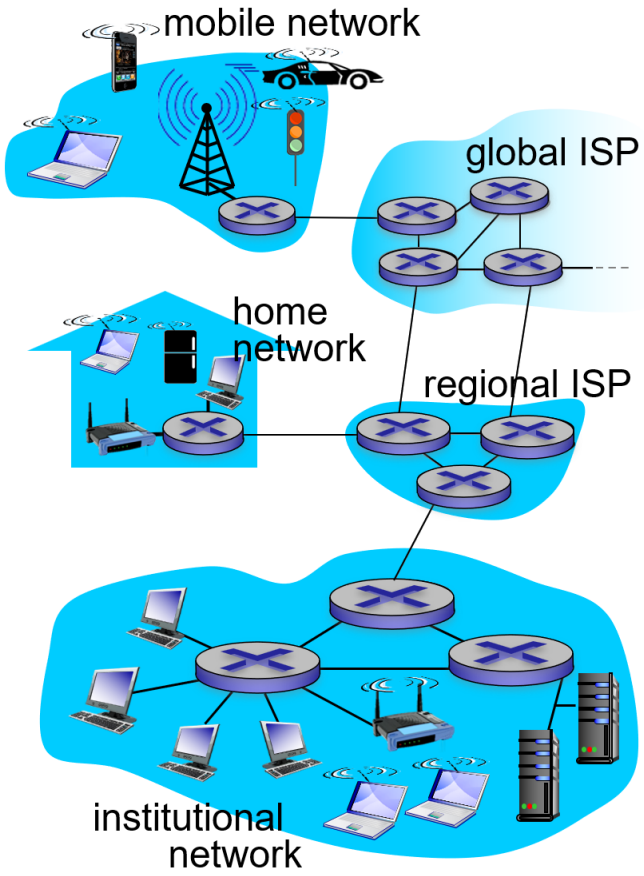
- *Internet:* "network of networks"
  - Interconnected ISPs
- *protocols* are *everywhere*
  - control sending, receiving of messages
  - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4/5G, Ethernet
- *Internet standards*
  - RFC: Request for Comments
  - IETF: Internet Engineering Task Force



Slide from Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross

A closer look at Internet structure

- *network edge:*
  - hosts: clients and servers
  - servers often in data centers
- *access networks, physical media:* wired, wireless communication links
- *network core:*
  - interconnected routers
  - network of networks



Slide from Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross Introduction 1-11

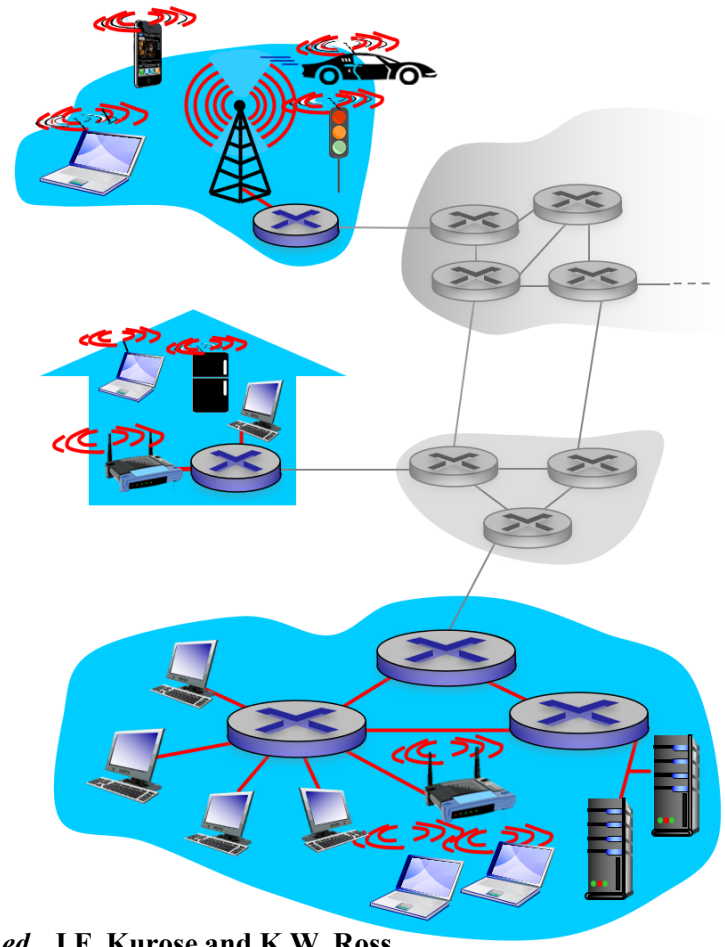
Access networks and physical media

Q: How to connect end systems to edge router?

- residential access nets
- institutional access networks (school, company)
- mobile access networks

keep in mind:

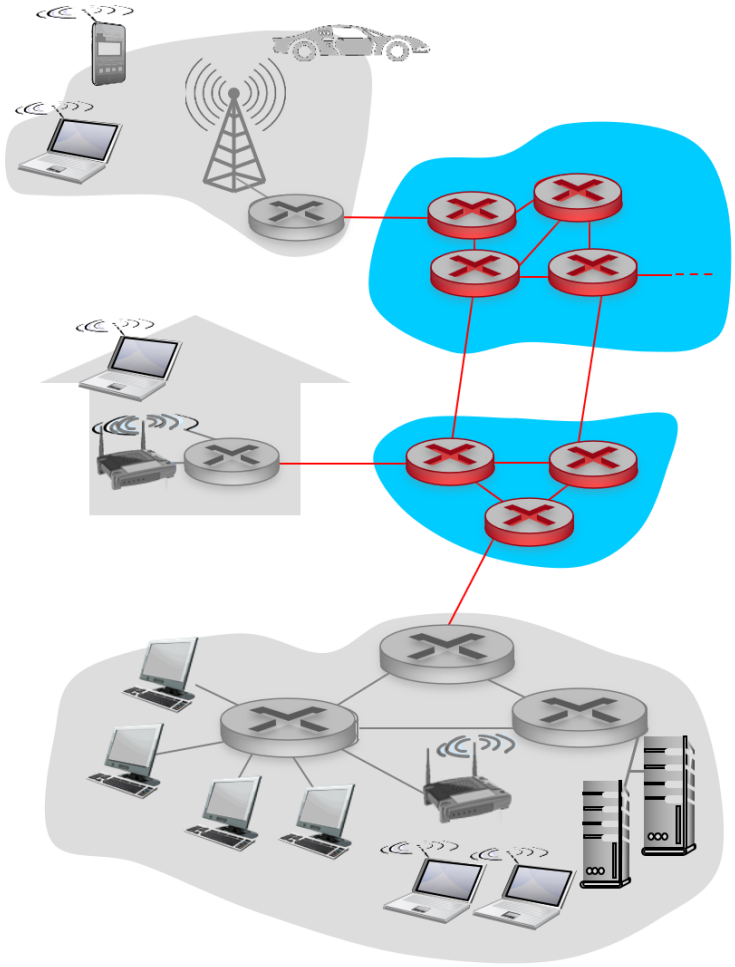
- bandwidth (bits per second) of access network?
- shared or dedicated?



Slide from Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross

The network core

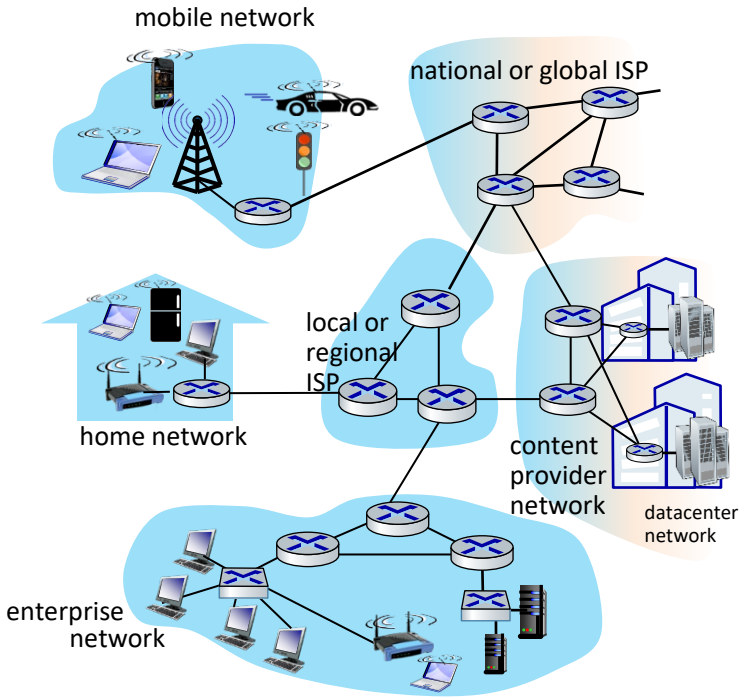
- mesh of interconnected routers
- packet-switching: hosts break application-layer messages into *packets*
  - forward packets from one router to the next, across links on path from source to destination
  - each packet transmitted at full link capacity



Slide from Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross

Internet structure: a "network of networks"

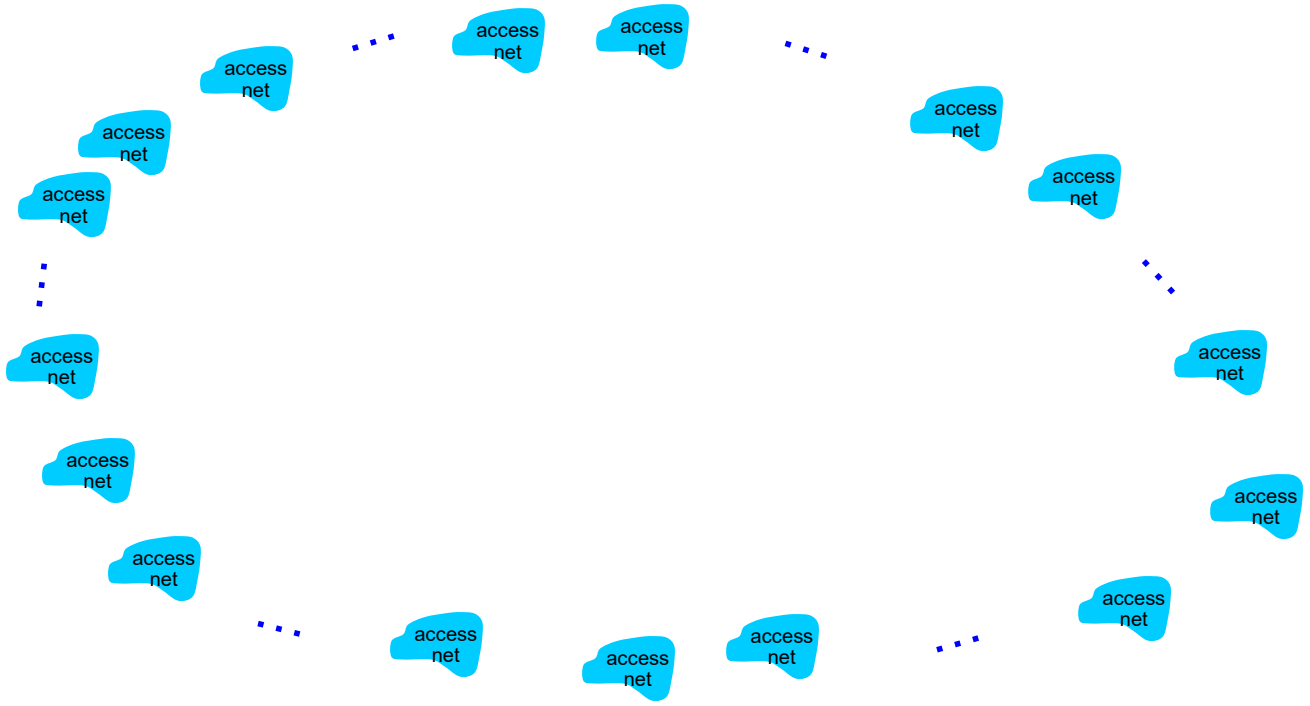
- hosts connect to Internet via access Internet Service Providers (ISPs)
- access ISPs in turn must be interconnected
  - so that *any* two hosts *(anywhere!)* can send packets to each other
- resulting network of networks is very complex
  - evolution driven by economics, national policies



Let's take a stepwise approach to describe current Internet structure

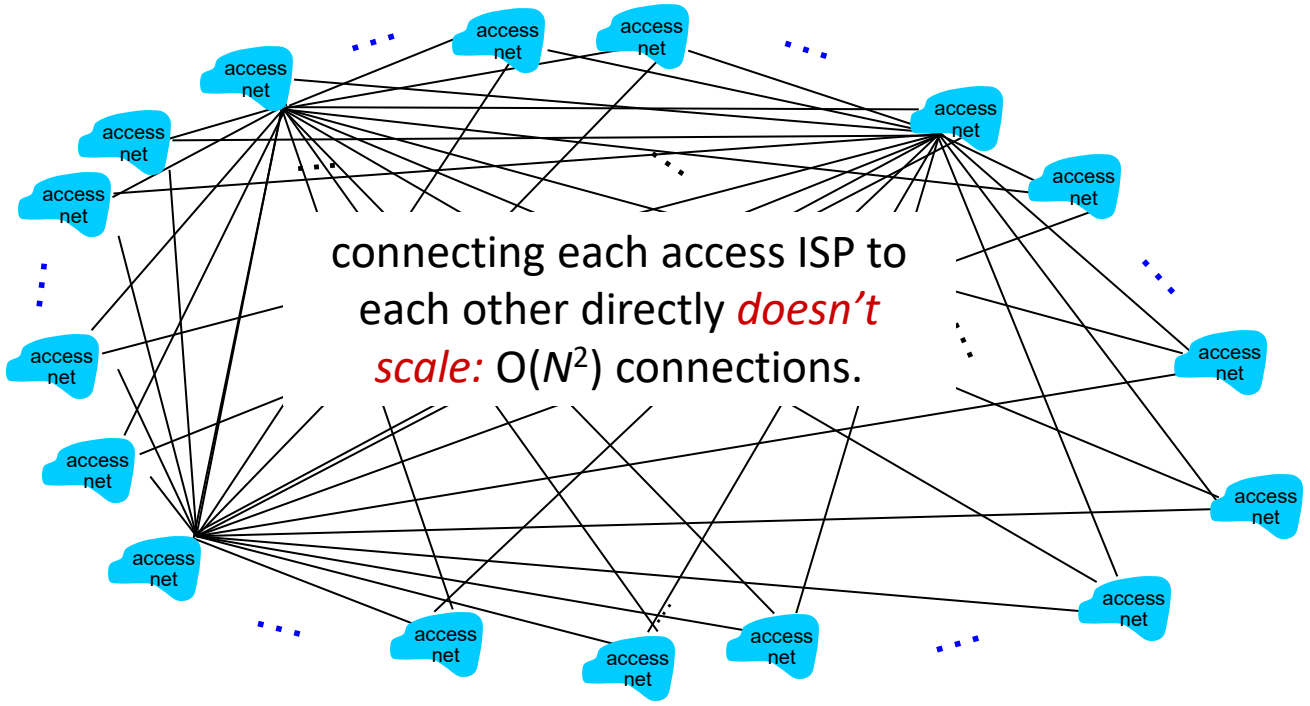
Internet structure: a “network of networks”

Question: given *millions* of access ISPs, how to connect them together?



Internet structure: a "network of networks"

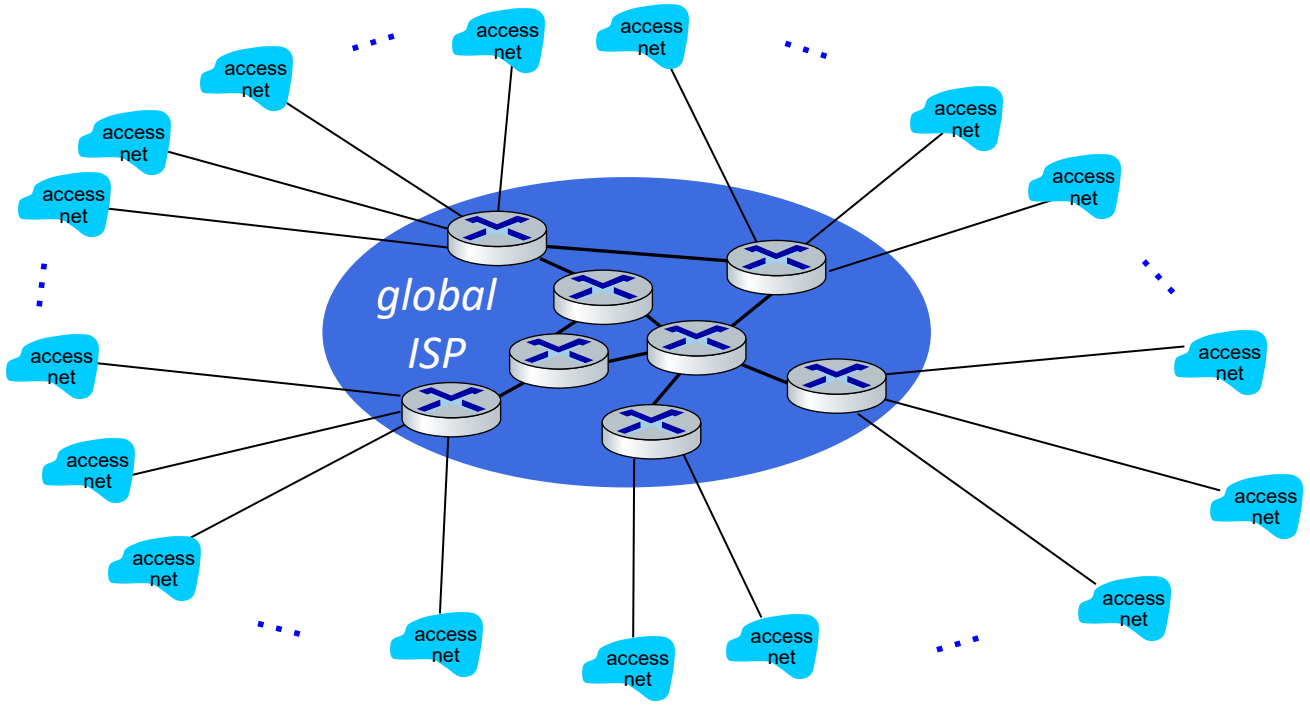
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Internet structure: a "network of networks"

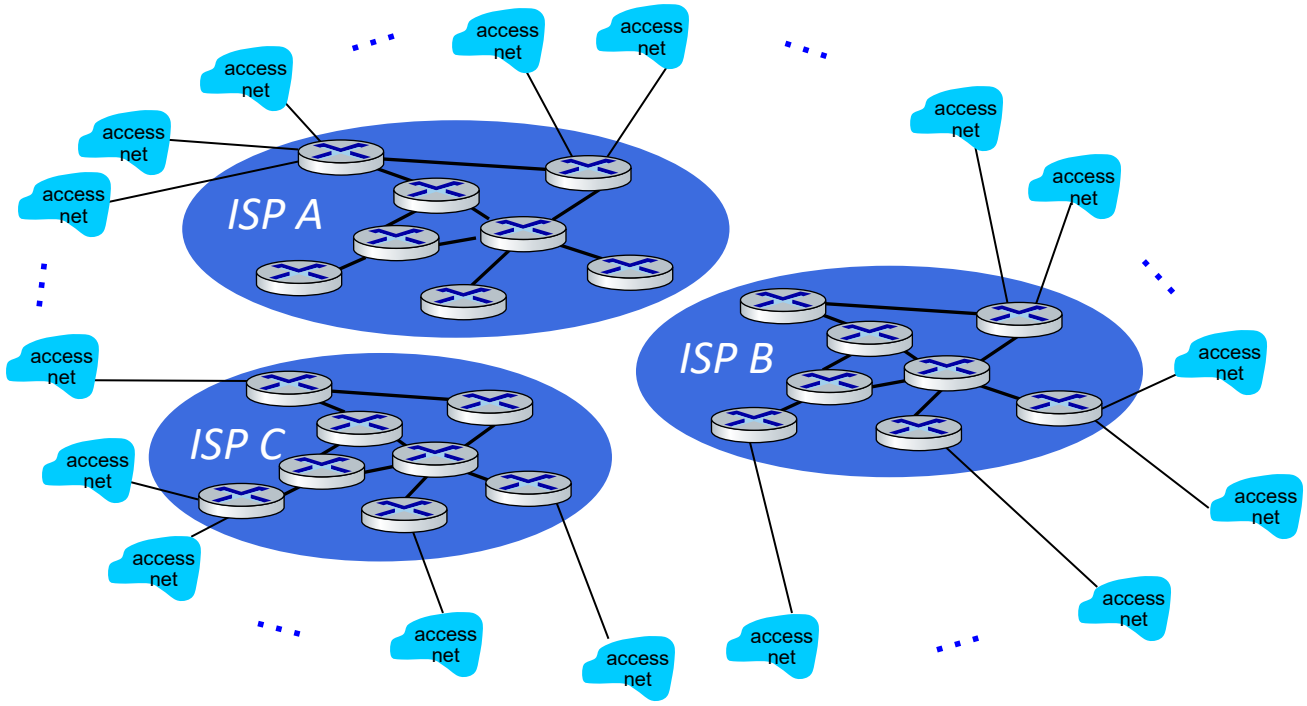
Option: connect each access ISP to one global transit ISP?

Customer and provider ISPs have economic agreement.



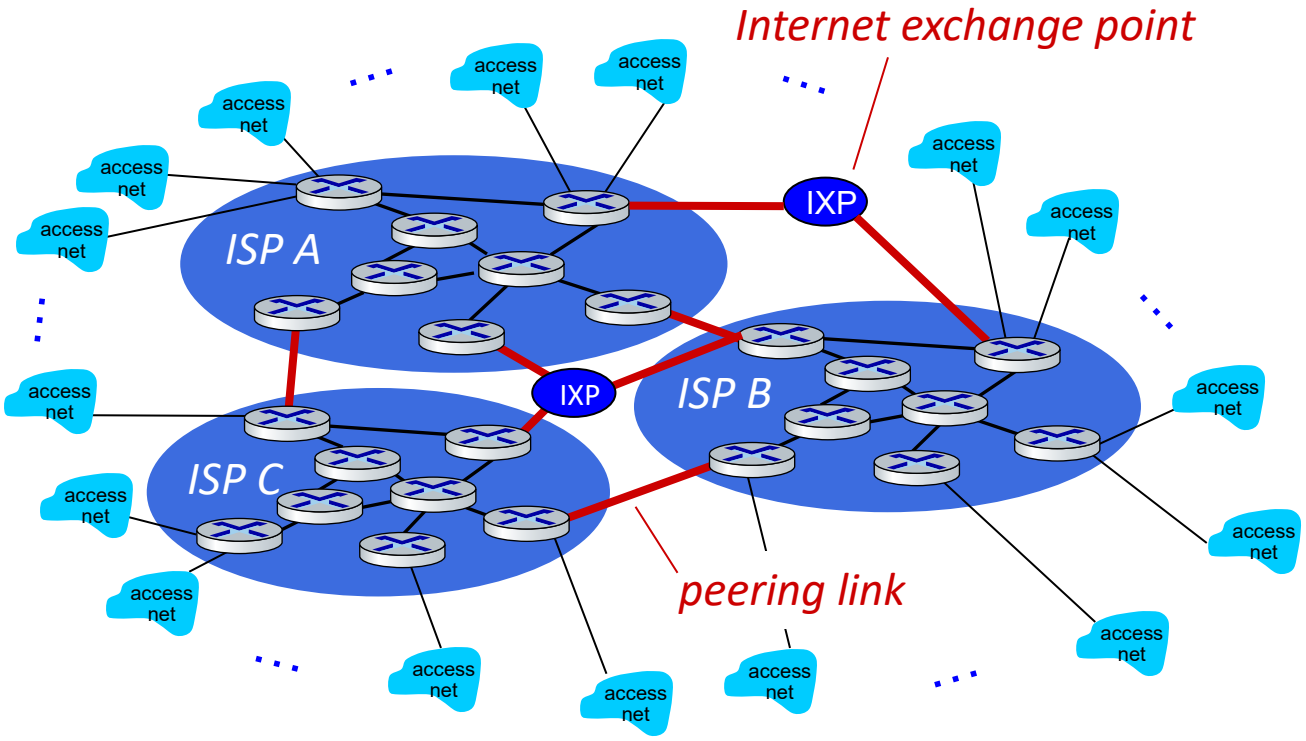
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors ….



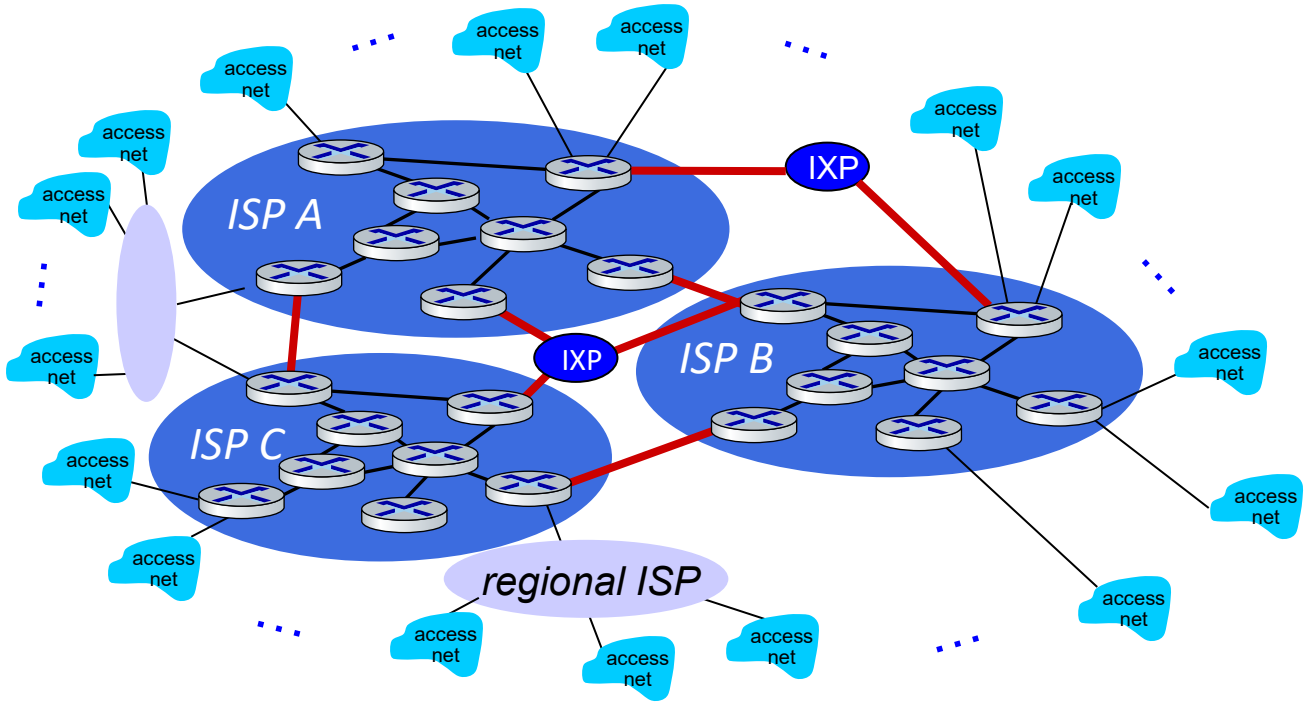
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors …. who will want to be connected



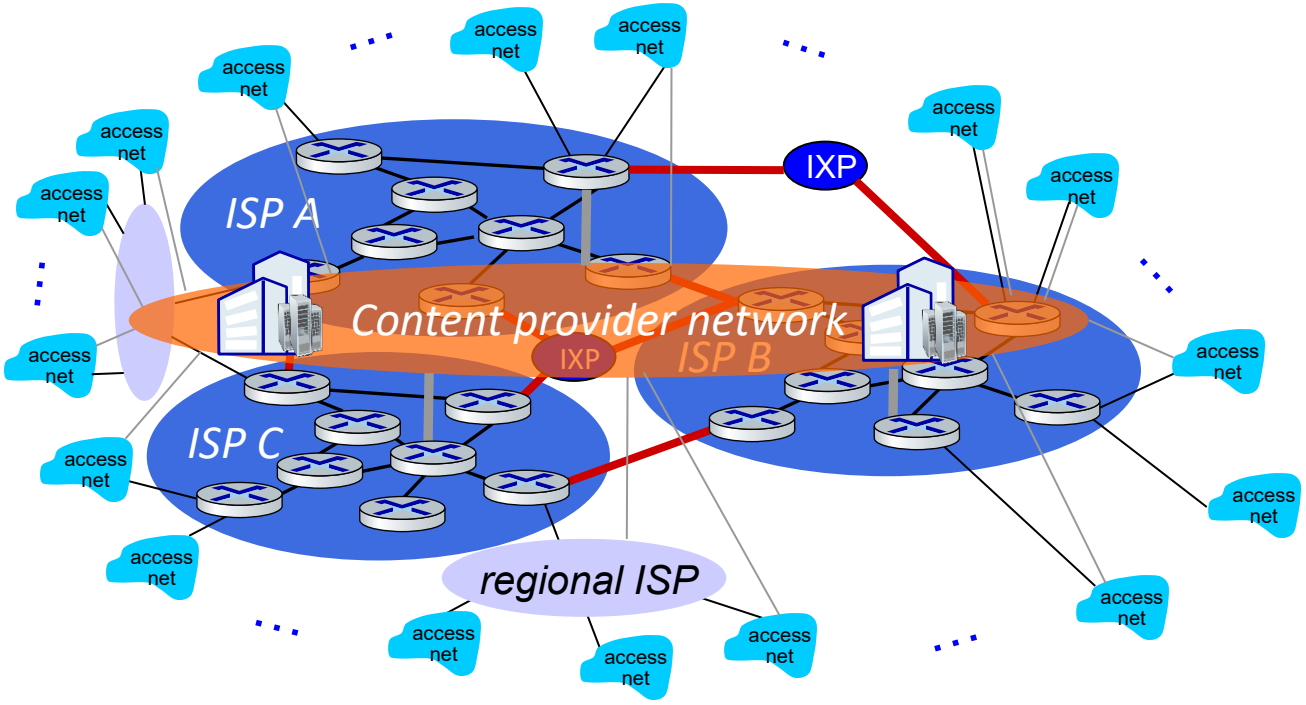
Internet structure: a "network of networks"

… and regional networks may arise to connect access nets to ISPs



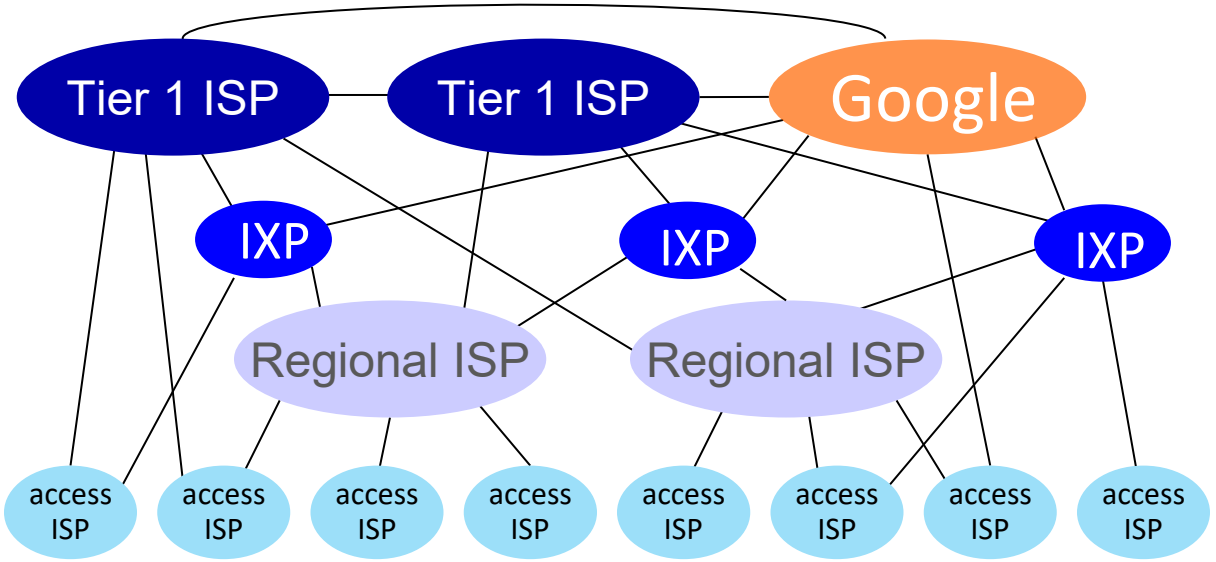
Internet structure: a "network of networks"

… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



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Internet structure: a "network of networks"

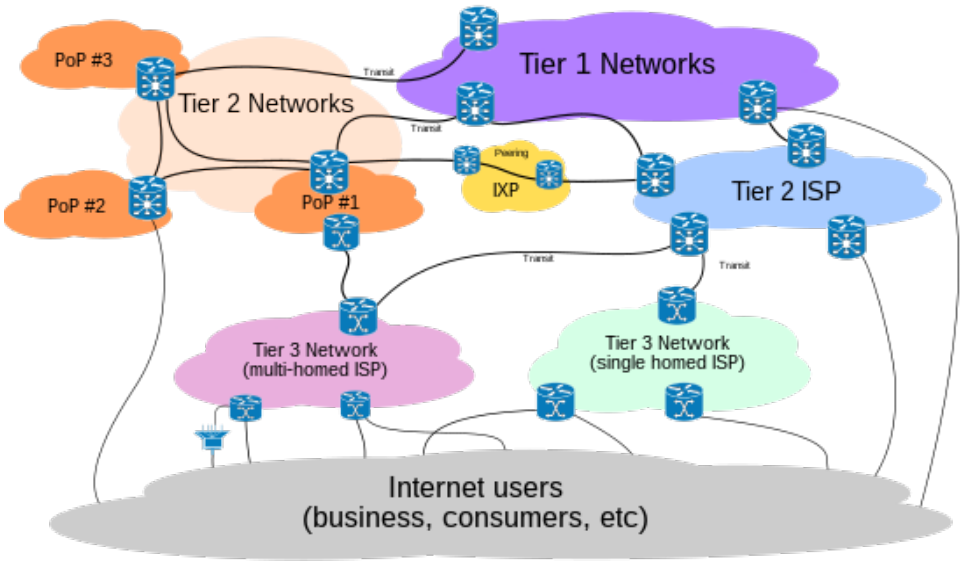


At "center": small # of well-connected large networks

- "tier-1" commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- content provider networks (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Network Operators Categories

- Tier 1
  - Global operators, Only peering, does not have provider
- Tier 2
  - Has peering with other ISP
  - Purchases transit IP from Tier 1
  - https://asrank.caida.org/
- Tier 3
  - solely purchases transit/peering from other networks to participate on the Internet
  - Single or multi-homed



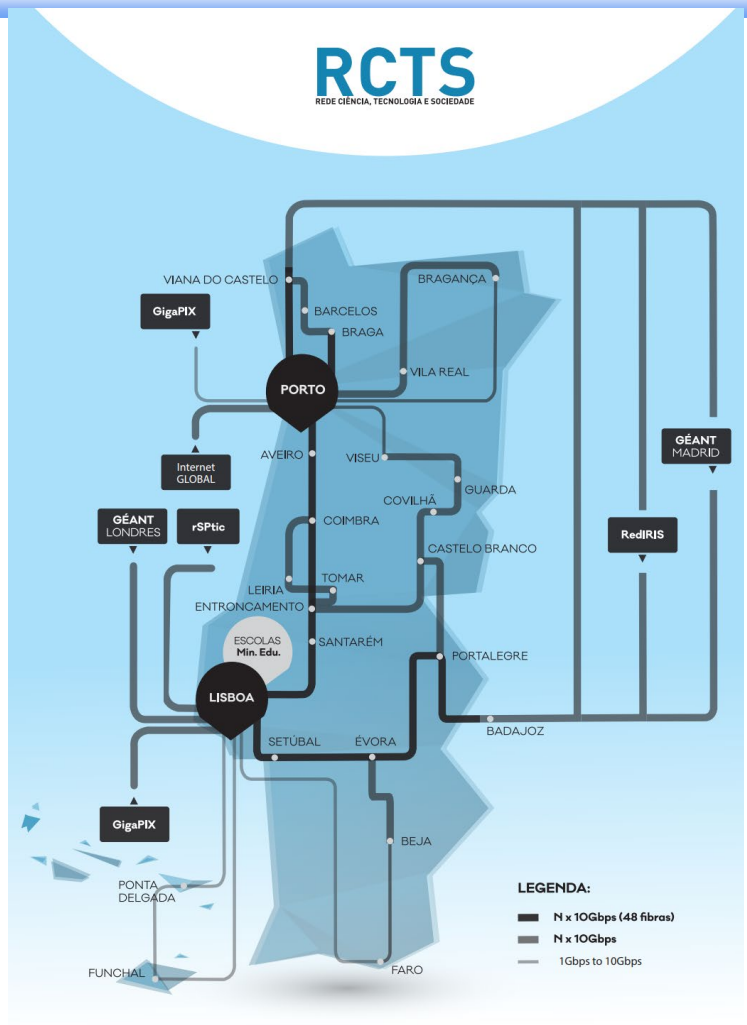
Tier 1 network - Wikipedia

Network Provider Categories

- Telecom Operators
  - Verizon (US), China Mobile, Deutsche Telekom, AT&T (US), ComCast (US), T-Mobile (US), NTT, China Telecom, Charter Communications, Vodafone, SoftBank (Japan), Telefónica (Spain), KDDI (Japan), BT (UK), Telstra, KT (Korea), BCE (Canada), Lumen (US), Tellus (Canada)
    - Cable, Landline, Mobile, Broadband, Digital TV, IPTV, Internet
- Global Network Operators (Tier 1 Network)
  - COGENT, LUMEN (Level 3), AT&T, NTT, Verizon, among others
- Private Network Operators
  - Google, Microsoft, Amazon
- ISP Service Providers
  - Focus on Internet Access, but not only
- National research and education network (NREN)
  - GÉANT (Europe backbone), FCCN (Portugal), RedIRIS (Spain), Internet2 (US), ESNet (US), RedClara (Latin America), RNP (Brasil), etc.

FCCN

- Science, Technology and Society Network (RCTS)
  - provide high-speed Internet connectivity and IT services to the Portuguese higher education and research system
  - Connection with other NRENS and other Networks
    - RedIRIS (Spain)
    - GÉANT (European NREN backbone)
    - GigaPIX
      - Portuguese Internet eXchange (PIX)
      - Allow several IP networks to interconnect in an efficient way (reduce latencies, for example)
    - Rede Operacional de Serviços Partilhados de Tecnologias de Informação e Comunicação da Administração Pública (rSPtic)

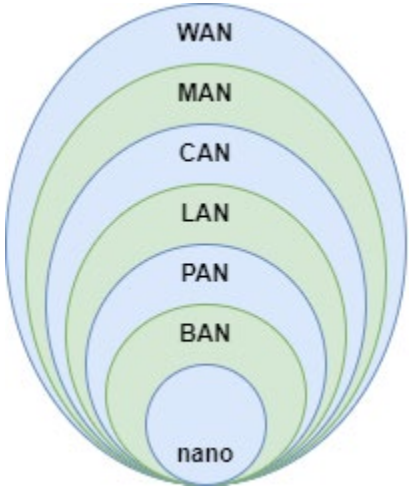


https://www.fccn.pt/media/2021/03/mapa-RCTS-fccn.pdf

www.fccn.pt

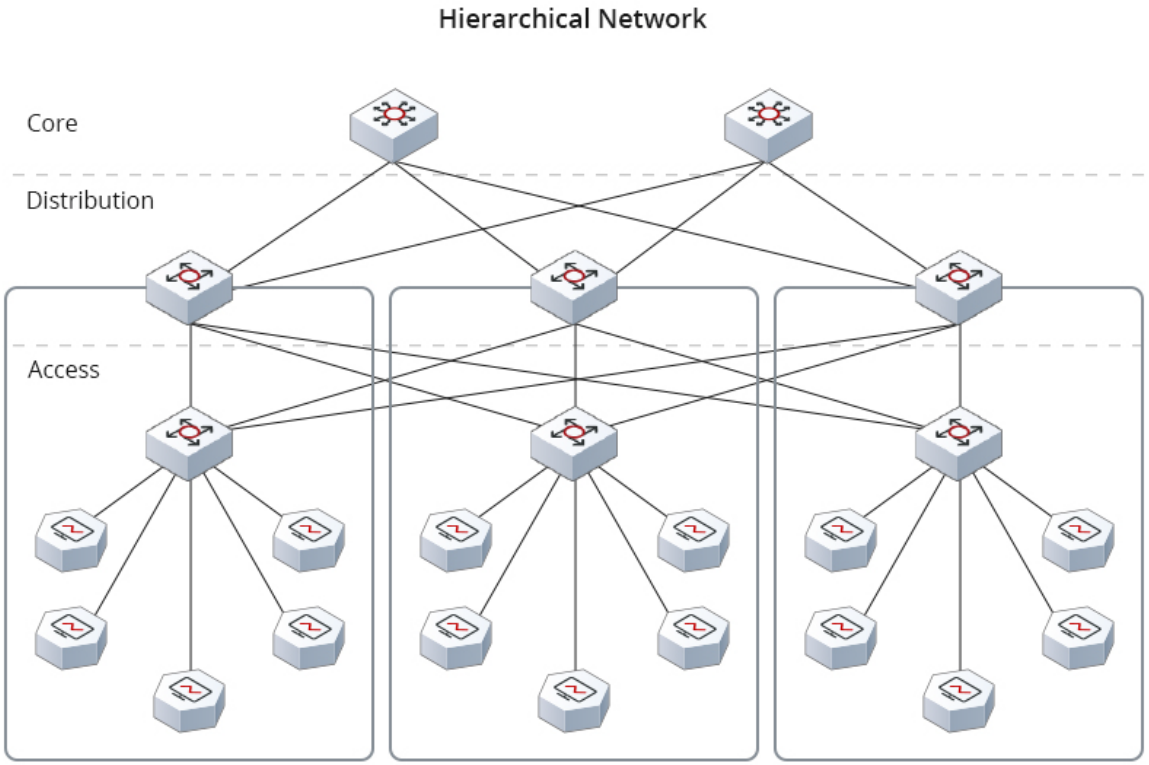
Networks Organization

- Classical design of networks
  - Can be applied to different types of networks
    - LAN, Campus Area Network (CAN), MAN, WAN
  - Also used by telecom operators
  - Comments below considers Telecom operators
- Access Networks
  - Provides the access to the customer (end-user = consumer or a business/enterprise)
  - Several technologies
    - Wired (copper and fiber) and Wireless
  - Evolution
- Distribution Networks
  - Also called aggregation network (or metro network)
  - Connection between the access and the core
  - Provides services such as routing and traffic management
- Core Networks
  - Also called backbone
  - High-speed
  - Reliability
  - Long-distance communication



Networks Organization
Local Area Network

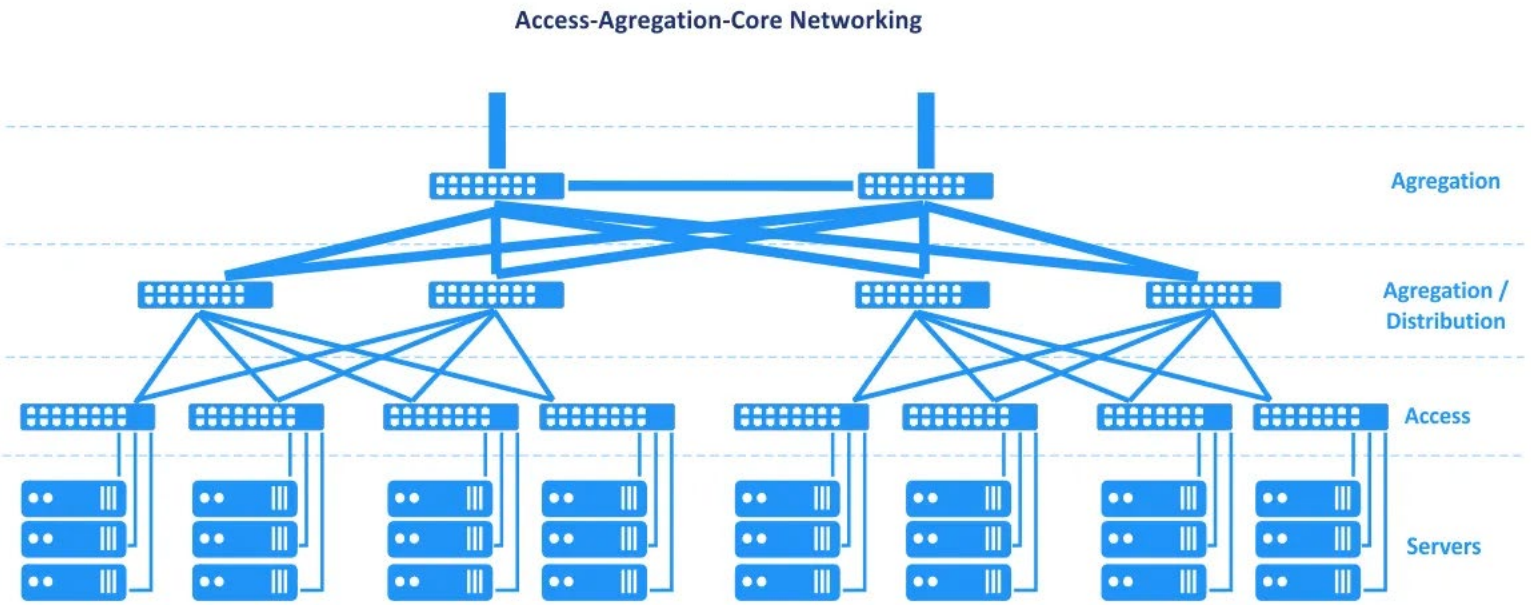
- Access, Distribution, Core
  - Inside an enterprise



Explore Hierarchical Networks: Access, Distribution, Core Layers | FS Community

Networks Organization
Local Area Network

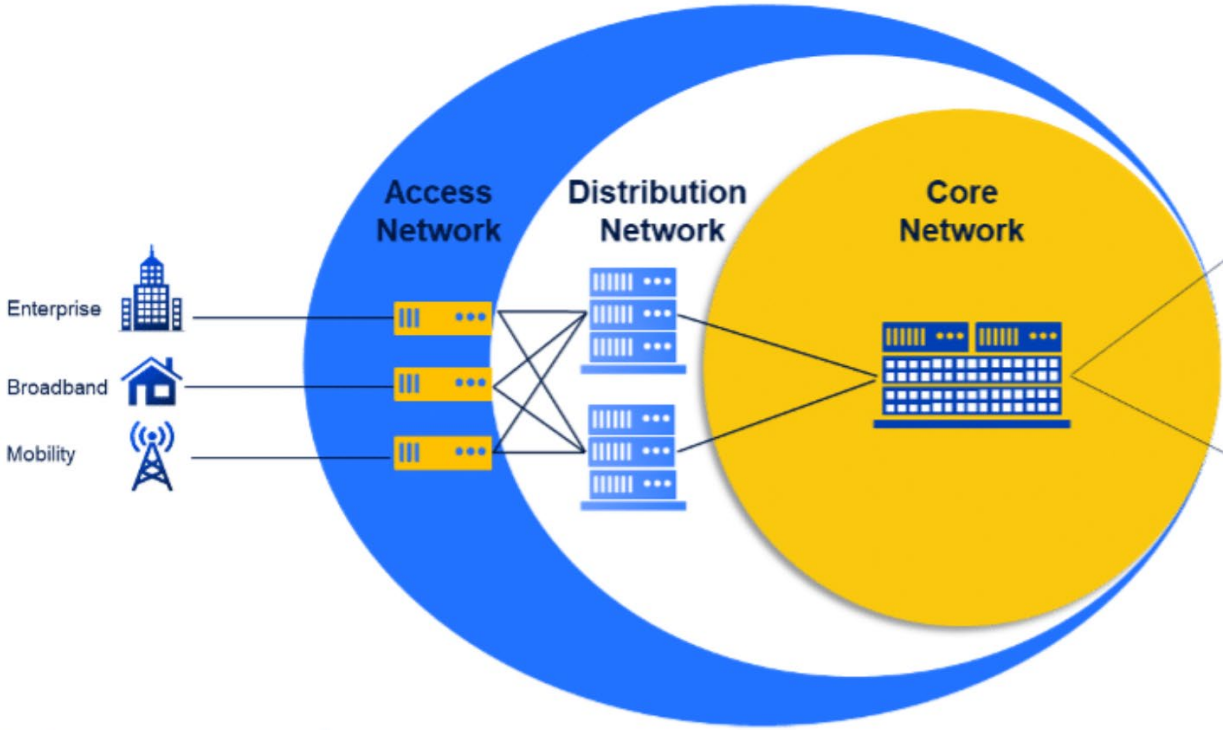
- Access, Distribution, Core
  - Inside a datacenter



https://www.nakivo.com/blog/msp-network-topology-for-beginners - 2023

Networks Organization
Telecom operator view

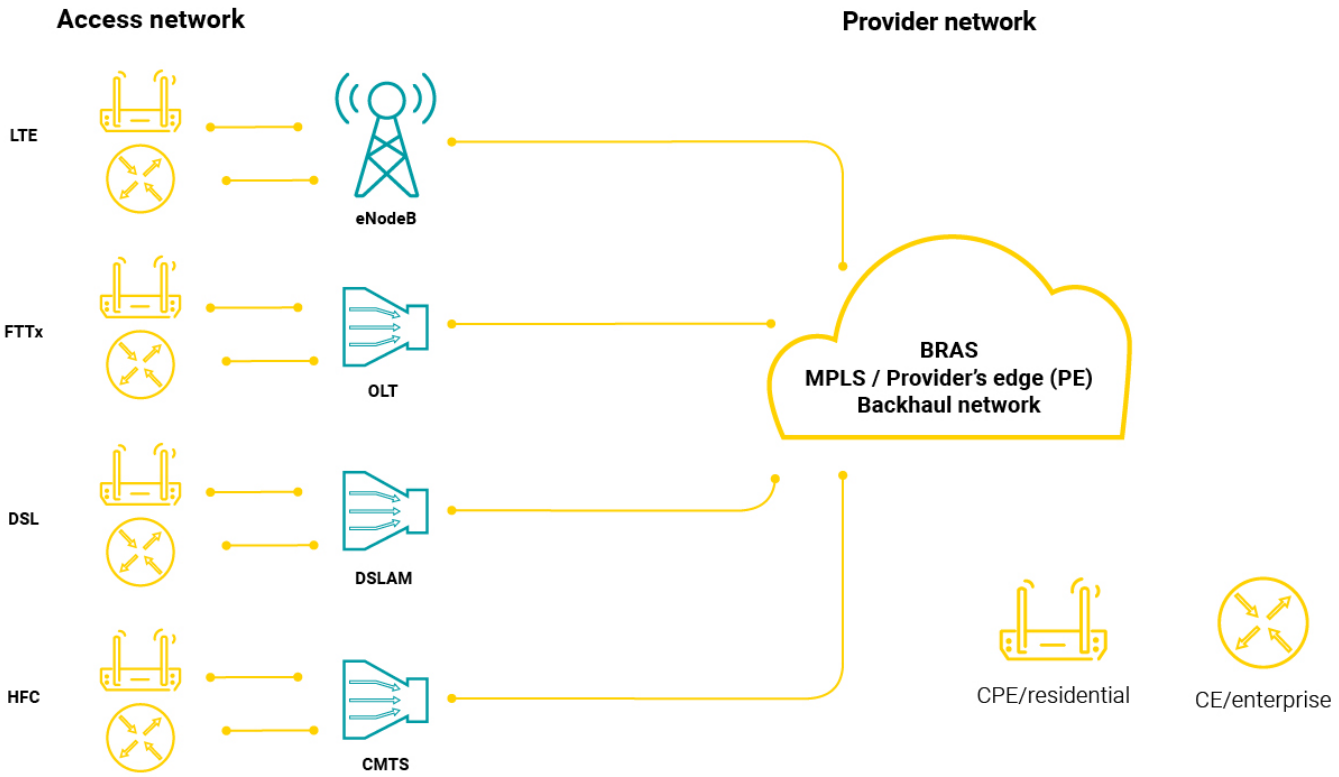
- Access, Distribution, Core



The Three layer Network Architecture

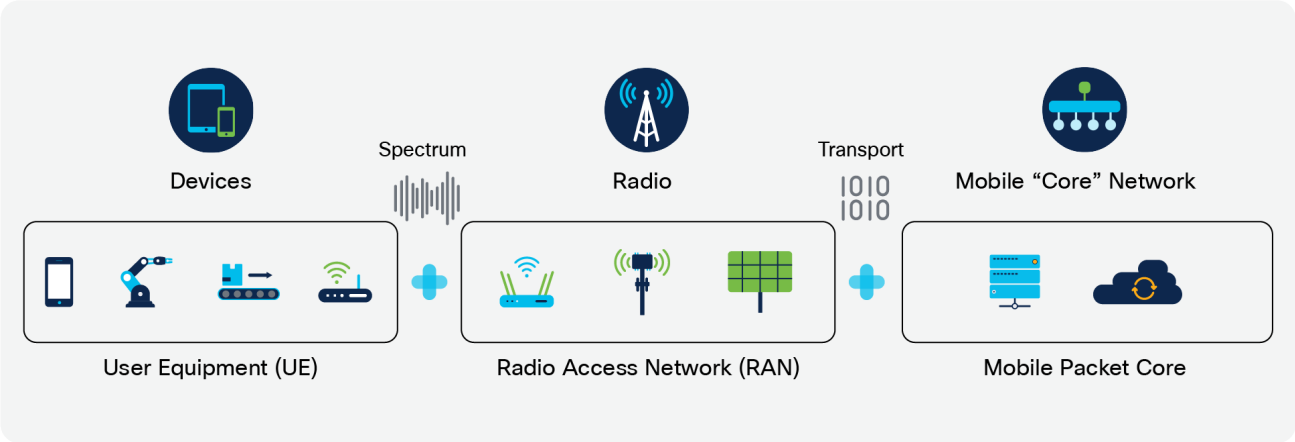
What is Core Network? A Deep Dive into Network Operations - DriveNets

Networks Organization
Telecom operator view



Networks Organization
Mobile operator view

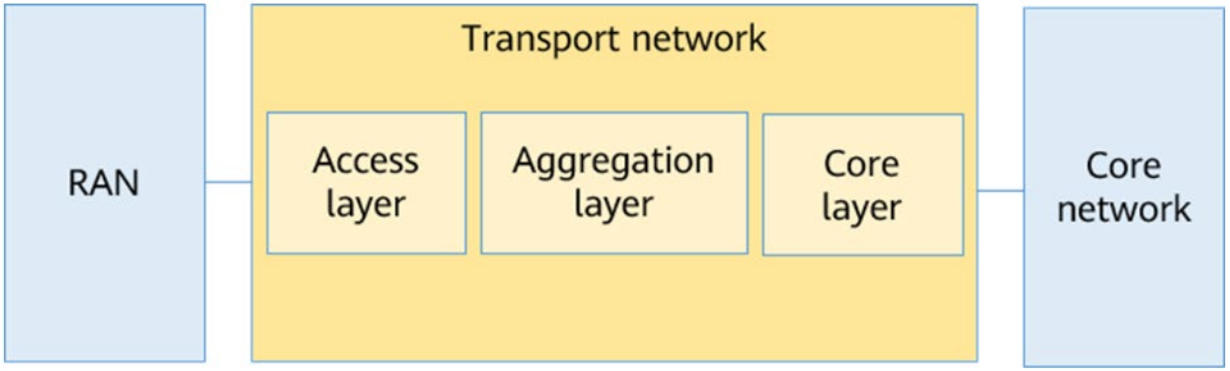
- Access
  - Called Radio Access Network (RAN)
- Distribution
  - Also called transport
  - Connects the RAN to the CORE
- Core
  - Called Mobile Core
  - Has its own organization based on several network functions (Virtual Network Functions)
    - Control Plane Functions
    - Data Plane Functions (or User Plane Functions)
    - Uses the Service-Based Architecture (SBA), based on HTTP/2 and REST APIs



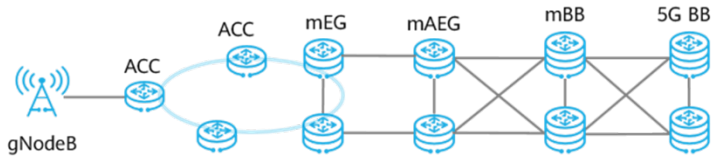
Products - 5G Enterprise Radio Access Networks White Paper - Cisco, 2021

Networks Organization
Mobile operator view

- RAN
- Distribution
  - Called here transport
  - Here also uses the three-layer organization (access, aggregation, core) in a recursive way
- Core



- Simplified view
  - ACC: access router
  - mEG: metro edge gateway
  - mAEG: metro aggregation edge gateway
  - mBB: metro backbone router
  - 5G BB: backbone router



What Is a 5G Transport Network? – Huawei, 2023

Telecom Operator Different Technologies

- In this case considers “last-mile” and “Access” the same meaning
- Considers legacy technologies
  - Digital Subscriber Line (DSL)
  - Digital Subscriber Line Access Multiplexer (DSLAM)
- Paper published in 2018
  - “Old” for the technology area
  - Some entities from 2G and 3G networks
    - 2G
      - Base Station Controller (BSC)
      - General Packet Radio Service (GPRS)
    - 3G
      - Radio Network Controller (RNC)
      - Serving GPRS Support Node (SGSN)
      - Gateway GRPS Support Node (GGSN)

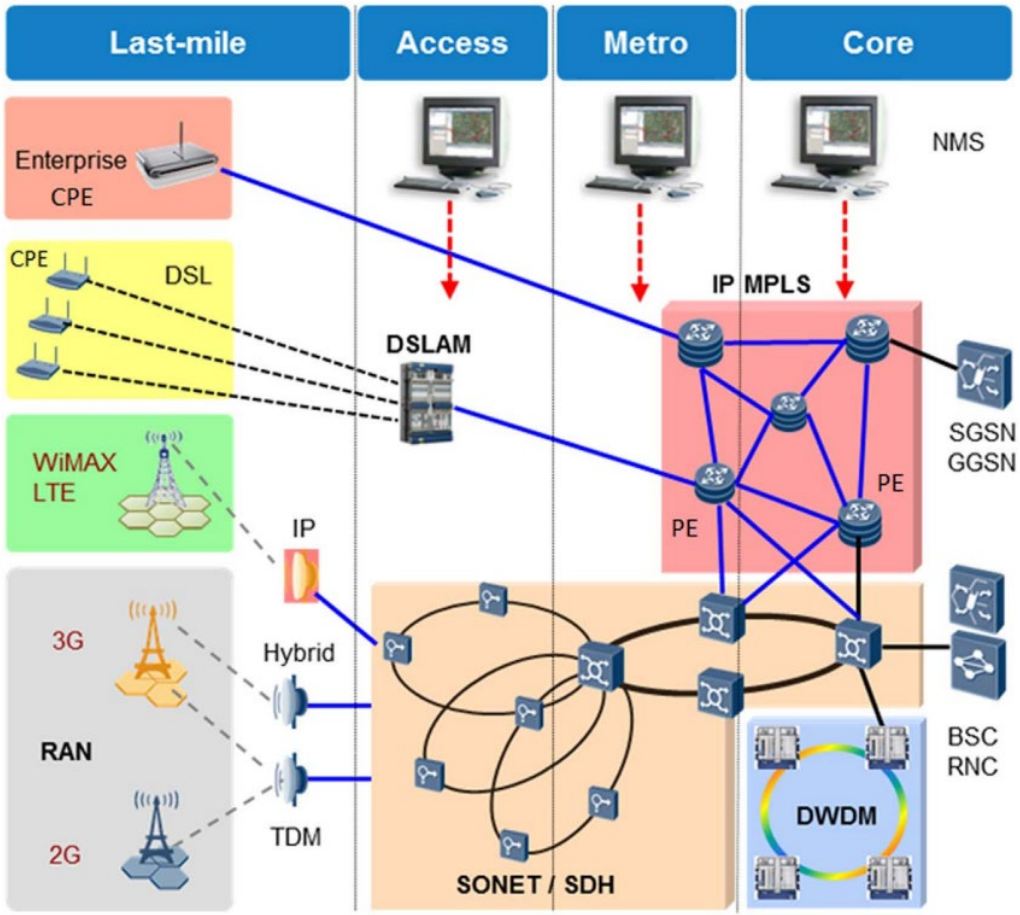


Fig. 1. *SP* legacy transport networks architecture.

Towards end-to-end integrated optical packet network: Empirical analysis, 2018

Telecom Operator Different Technologies

- Present newer technologies
  - FTTH (Fiber to the Home)
  - LTE (Long Term Evolution) a.k.a 4G
  - Software Defined Networking (SDN)
  - Network Function Virtualization (NFV)
- Paper published in 2018
  - Does not consider 5G!

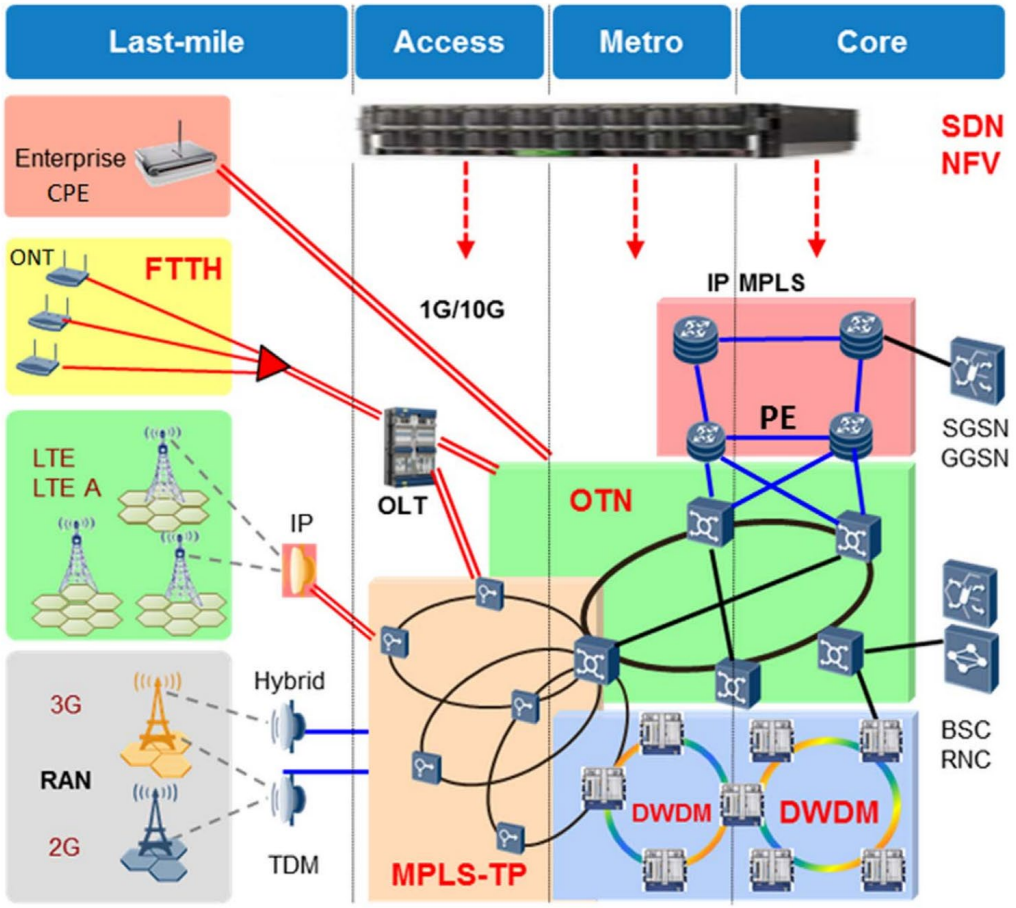


Fig. 4. Transport Network transformation towards E2IOPN.

Towards end-to-end integrated optical packet network: Empirical analysis, 2018

Access Networks, the Last Mile

- more bandwidth consumption
  - IPTV, Music Streaming, Video Contents, Web Browsing, Gaming,
- Increasing rates (average values – SpeedTest)
  - Download: 100 Mbps, 300 Mbps, …
  - Upload: 10 Mbps, 30 Mbps, 100 Mbps
- Reduced latency
  - Ultra reliable Low Latency Communications (URRLC) – 5G
- Reliability
  - Multi Link Operation (MLO) in Wi-Fi 7
- New Uses Cases
  - Extreme Worldwide Coverage; Industry of the Future; Large-Scale Digital Twins based Systems; Advanced Remote Interactions; Invisible Safe Zones
- New Verticals
  - Industry, Entertainment, Health,
- New Enablers

Networks Evolution Trends Will Affect Access

- Some Use Cases for Future Networks[1]
  - Extreme Worldwide Coverage
  - Industry of the Future
  - Large-Scale Digital Twins based Systems
  - Advanced Remote Interactions
  - Invisible Safe Zones
- Several New Enablers
  - Energy, Sensing and Actuation, Communication, Softwarization, Immutability (DLT), Intelligence (AI and related tech), Security, Quantum
- Also new sensors
  - Contributing to a fusion between physical and digital worlds



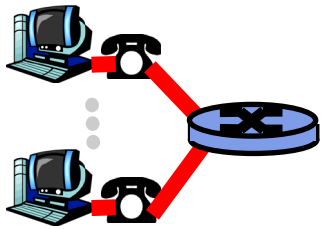
[1]A Detailed Relevance Analysis of Enabling Technologies for 6G Architectures, 2023
DOI - 10.1109/ACCESS.2023.3301811

Access Networks
Connectivity options

- Early days
  - Using telephone lines
    - Public switched telephone network (PTSN)
      - Conventional modems (19.2K, 38.4K, 57.6K)
    - Integrated Services Digital Network (ISDN) - 64K+64K
- xDSL - x Digital Subscriber Line
- Cable
  - Hybrid fiber-coaxial (HFC)
- Fixed wireless: microwave/laser
- FTTx
  - Fiber to the home/Curb/Building/Room
- Satellite
  - LEO, MEO, GEO
- Mobile Networks
  - GSM/3G, 4G (LTE), 5G
- Wireless Networks
  - Wireless Lan(IEEE 802.11x)
  - Bluetooth (IEEE 802.15.1) (PAN)
  - Body Area Network (IEEE 802.15.6)

Residential access: point to point access

- Dialup via modem
  - up to 56Kbps direct access to router (often less)
  - Can't surf and phone at same time: can't be "always on"
- DSL: digital subscriber line
  - deployment: telephone company (typically)
  - around several Mbps upstream
  - around tenths of Mbps downstream
  - dedicated physical line to central office



xDSL: Digital Subscriber Line

- Use existing twisted pair copper plant
  - point-to-point link
  - Frequencies (0-4 kHz) for voice calls (POTS – Plain Old Telephone service, based on analog signals).
  - Higher Frequencies (starting at 25 kHz) for data transmission.
- But not a great transmission medium
  - single pair, long, gauge & material changes
  - high frequency loss, bridge taps and load coils
- interference sources
  - RF pickup/egress, thermal noise, reflections
  - Near End crosstalk (NEXT), Far End Crosstalk (FEXT)
  - Digital Signal Processing (DSP) can mitigate the problem

xDSL variants

- HDSL: 1.5Mbps, symmetric, 2 pair, no POTS, up to 4Km
- SDSL: 1.5Mbps, symmetric, 1 pair, up to 6Km
- ADSL: 640-8Mbps down, 64-800kbps up, 1 pair, POTS/ISDN, up to 6Km
- VDSL: 6-52Mbps down, 26Mbps up, 1pair, 1.5Km
  - also 1,2,4,6,8,12Mbps symmetric
- VDSL2: 100Mbps up / down, 1pair, 3Km
- Bandwidth negotiation and noise monitoring
- Asymmetric variants to reflect current traffic patterns

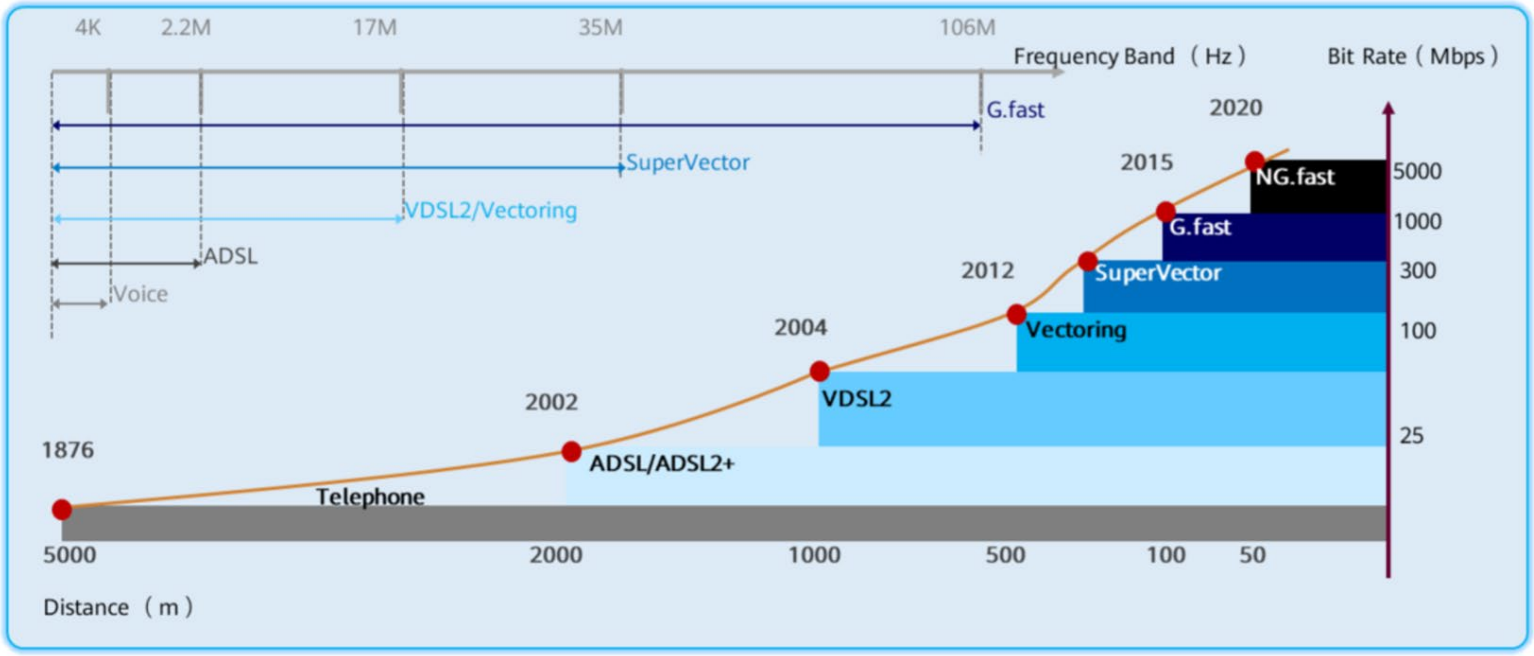
xDSL Comparison

Standard	Max Download Speed	Max Upload Speed	Max Distance from Central Office	Frequency Range	Year of Release	ITU-T Standard Number	Usage
ADSL	8 Mbps	1 Mbps	~5.5 km (~18,000 ft)	25 kHz – 1.1 MHz	1999	G.992.1	General internet, home use
ADSL2	12 Mbps	1.3 Mbps	~5.5 km (~18,000 ft)	25 kHz – 1.1 MHz	2002	G.992.3	Improved speed and efficiency
ADSL2+	24 Mbps	3.3 Mbps	~3.7 km (~12,000 ft)	25 kHz – 2.2 MHz	2003	G.992.5	High-speed internet for homes
SDSL	2.3 Mbps (symmetric)	2.3 Mbps (symmetric)	~3 km (~10,000 ft)	25 kHz – 2.2 MHz	1998	G.991.2	Business-grade symmetric internet
VDSL	52 Mbps	16 Mbps	~1.5 km (~5,000 ft)	25 kHz – 12 MHz	2001	G.993.1	High-speed home and business
VDSL2	100 Mbps	100 Mbps	~1.6 km (~5,200 ft)	25 kHz – 30 MHz	2005	G.993.2	Ultra-high-speed services
G.fast	1 Gbps	1 Gbps	~100 meters (~330 ft)	Up to 106 MHz	2014	G.9701	Fiber-like speeds over short distances

- ADSL - Asymmetric, download speeds are higher than upload speeds
- SDSL - Symmetric, equal download and upload speeds
- VDSL/VDSL2: Faster versions with a shorter range, often used for high-definition streaming or business use
- G.fast - newer standard offering gigabit speeds but limited to very short distances

xDSL Technology

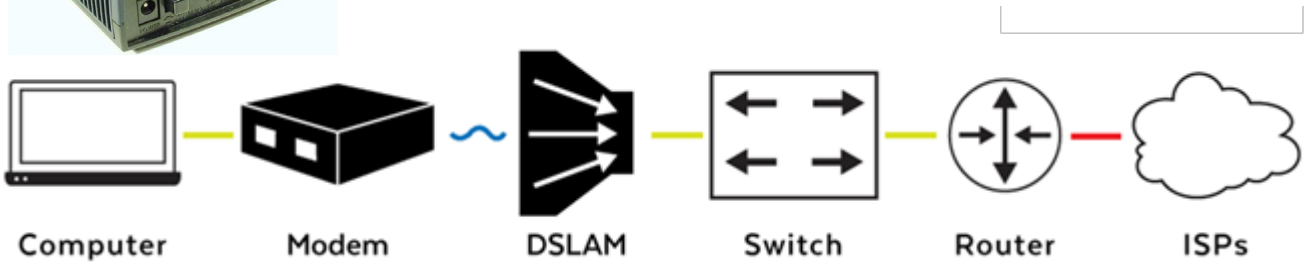
- Bandwidth and Effective Transmission Distance

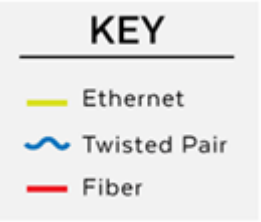


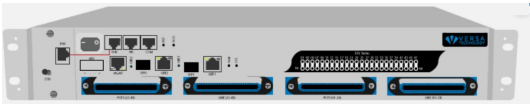
Digital Subscriber Line Access Multiplexer (DSLAM)

- Network device used to aggregate and manage multiple DSL connections from individual subscribers into a high-speed backbone
- Multiplexes (combines) all these individual DSL connections into a single, higher-capacity data stream
- Separates the high-frequency data traffic (internet) from the low-frequency voice traffic
- Converts the incoming DSL signals into IP packets or Ethernet frames









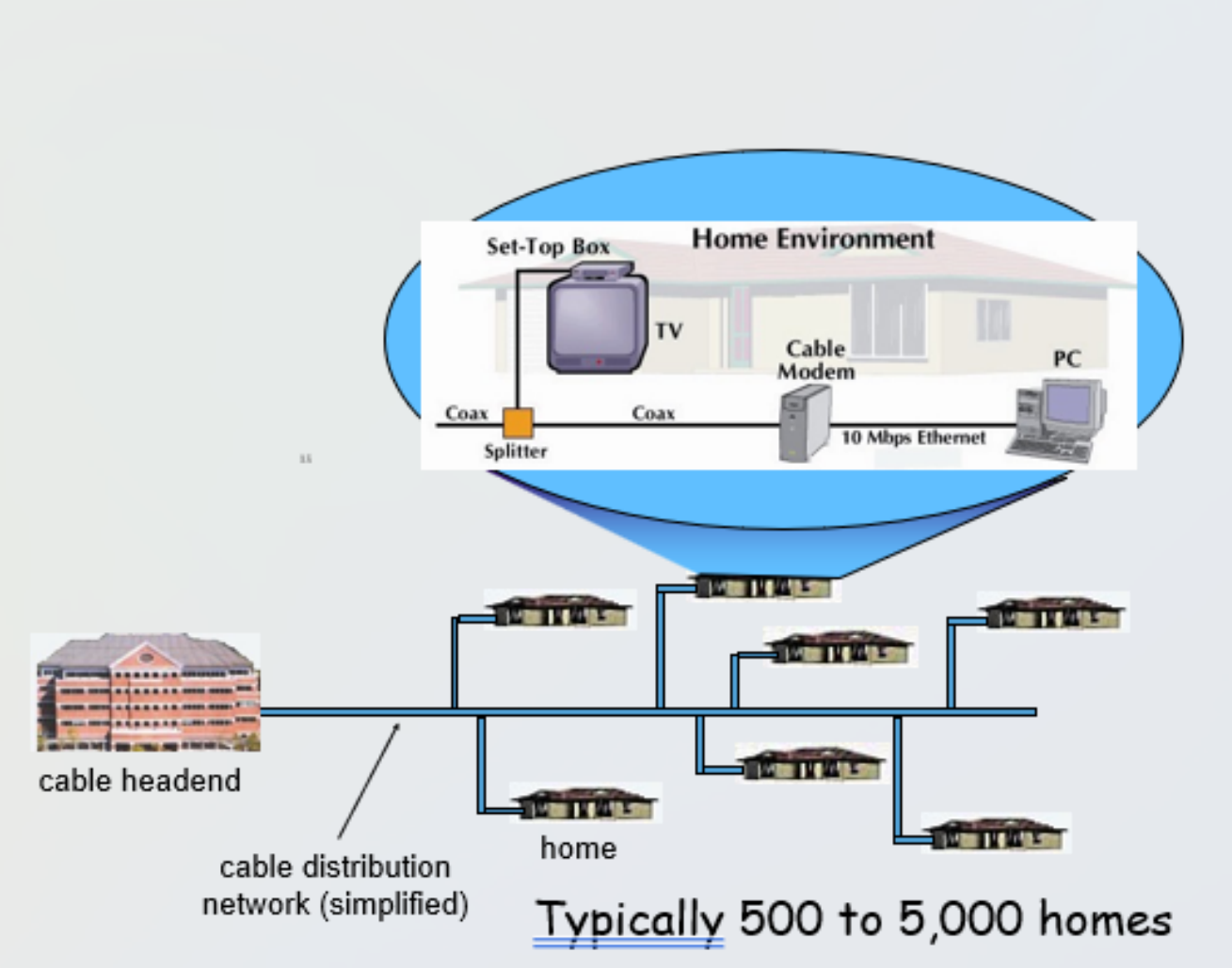
What is DSLAM?

Cable Modems

- Uses CATV coax tree from Head End
  - serve 1000's of customers
  - rapid rollout -- can split tree later
- 30-40 Mb/s shared downstream bw
- single 6MHz channel (same as a TV station)
  - 64/256 QAM encoding
  - head-end scheduled



Cable Network Architecture: Overview



Cable Modems

- Upstream channel is harder (10-100Mbps)
  - 16 QAM - 64 QAM (Quadrature Amplitude Modulation)
  - need MAC protocol for Collision Detect and retransmission, fair bandwidth sharing
  - large distances require ranging optimizations
- Encryption necessary for both channels
  - DES block cipher



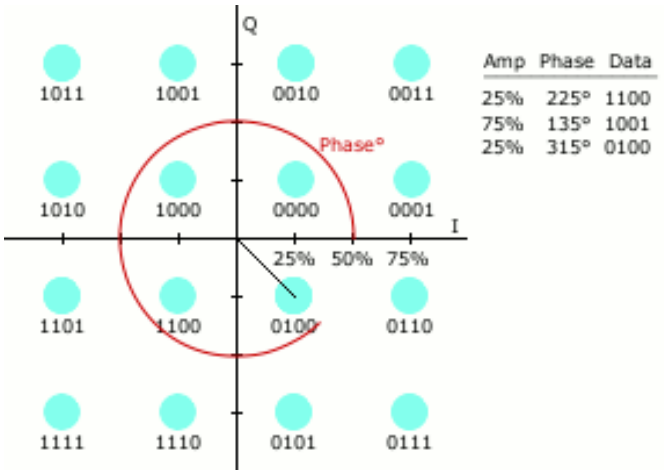
Data Over Cable Service Interface Specification (DOCSIS)

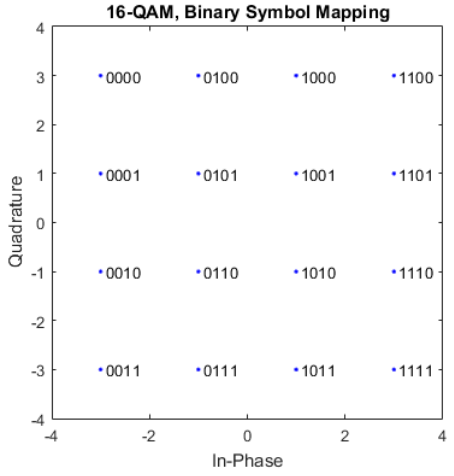
- Standard about high-speed internet services over existing cable television (CATV) infrastructure
- Allows cable companies to offer broadband internet access, voice services (VoIP), and even IPTV services alongside traditional cable TV

DOCSIS Version	Maximum Downstream Speed	Maximum Upstream Speed	Downstream Channels	Upstream Channels	Modulation Scheme	Release Year
DOCSIS 1.0	40 Mbps	10 Mbps	Single Channel (6 MHz)	Single Channel (6 MHz)	64-QAM (downstream), QPSK (upstream)	1997
DOCSIS 1.1	40 Mbps	10 Mbps	Single Channel (6 MHz)	Single Channel (6 MHz)	64-QAM, Support for QoS	1999
DOCSIS 2.0	40 Mbps	30 Mbps	Single Channel	Single Channel	256-QAM (downstream), 64-QAM (upstream)	2001
DOCSIS 3.0	1 Gbps (1,000 Mbps)	200 Mbps	Channel Bonding (up to 32 downstream)	Channel Bonding (up to 8 upstream)	256-QAM (both)	2006
DOCSIS 3.1	10 Gbps	2 Gbps	OFDM (up to 32 downstream)	OFDMA (up to 8 upstream)	4096-QAM (both)	2013
DOCSIS 4.0	10 Gbps	6 Gbps	OFDM	OFDMA	4096-QAM	2017

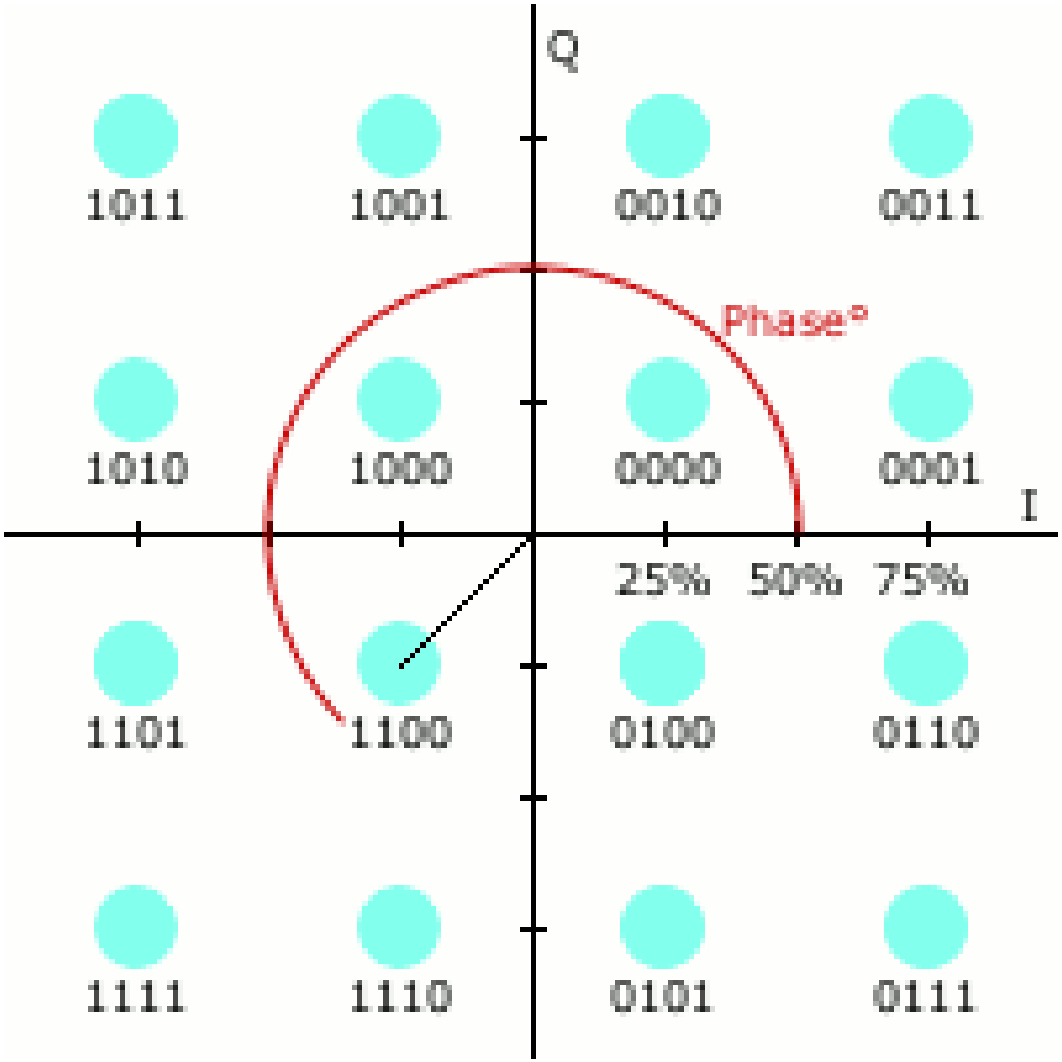
16-QAM Modulation

- Uses two orthogonal carriers (usually sine and cosine)
  - combines phase modulation and amplitude modulation
- Modulates the Amplitude of these two carriers
- Uses a constellation diagram to represent the modulated signal
  - Two dimensional xy-plane
  - maps data modulation information (including amplitude and phase information about signals) to polar coordinates.
- QAM is used in digital communications systems (Cable Modem, 802.11 Wi-Fi standards, etc.)





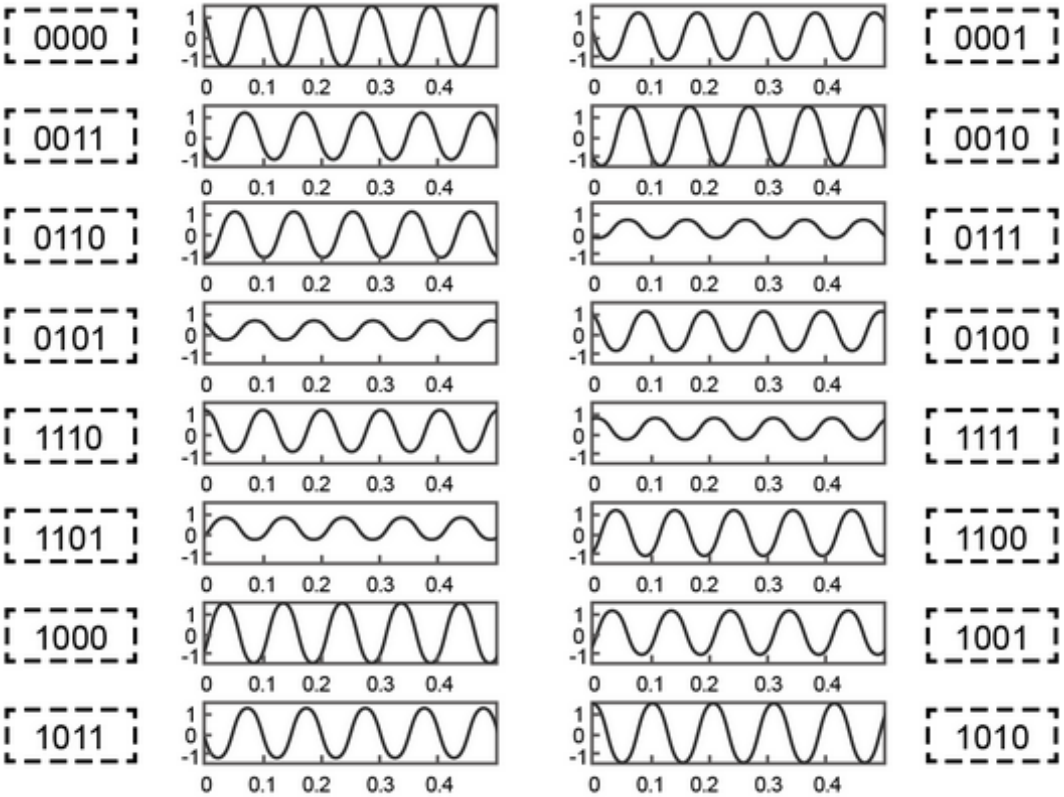
16-QAM example



Amp	Phase	Data
25%	225°	1100

16-QAM Modulation

- 16-QAM modulate symbols into 16 different waveforms
- Total of 16 types of symbols are available, each of which may transmit 4-bit information.



Notice: for each symbol (4 bits/symbol) the same frequency is used

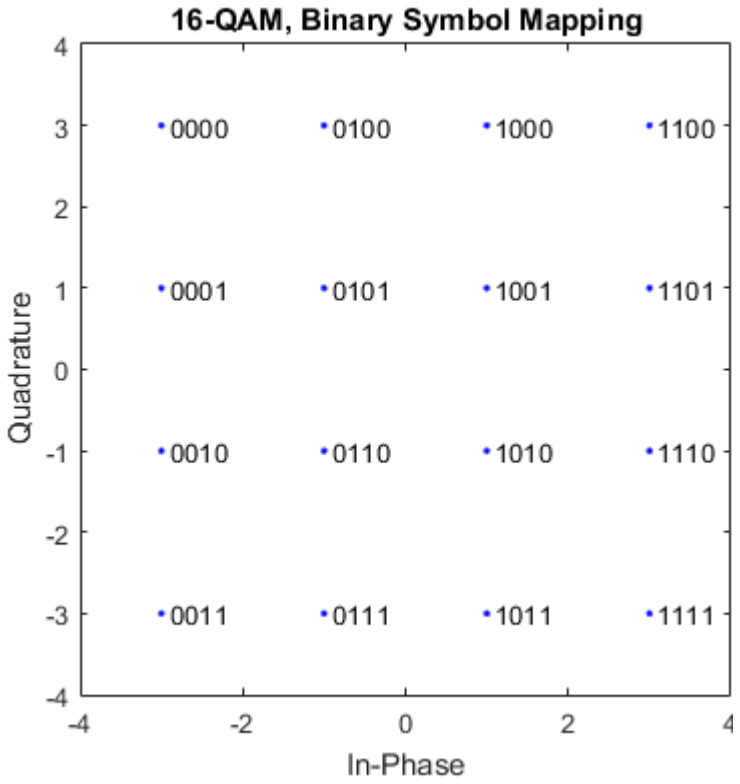
QAM Modulation Gains

- The number of constellation points in the grid determines the number of bits carried in each symbol.

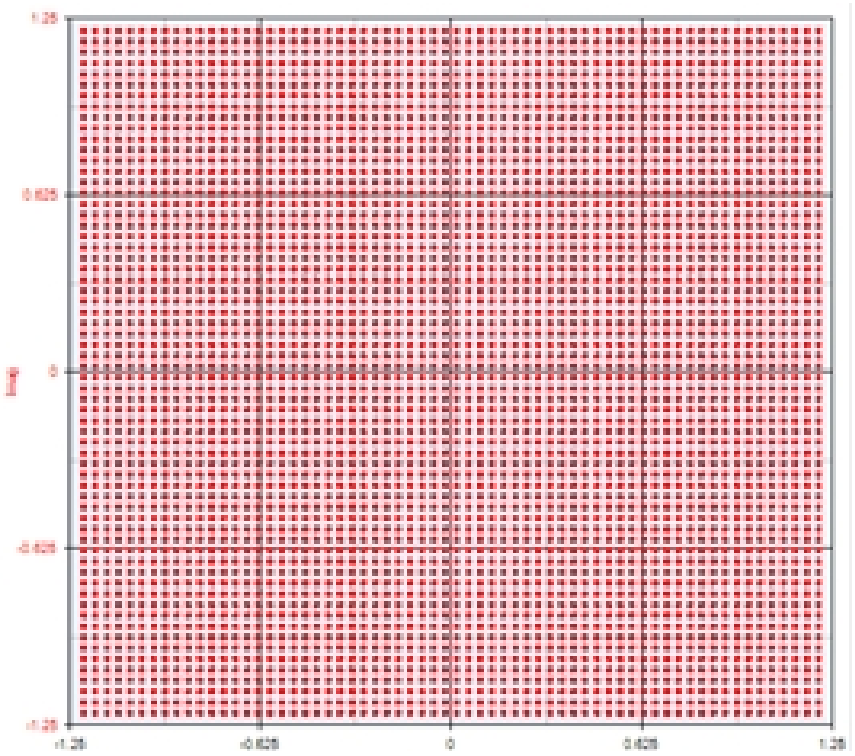
Modulation Level (QAM)	Bits/Symbol Bits/s/Hz
QAM-8	3
QAM-16	4
QAM-32	5
QAM-64	6
QAM-128	7
QAM-256	8
QAM-512	9
QAM-1024	10
QAM-2048	11
QAM-4096	12

Other QAM Modulations Modulation Symbols

16-QAM Modulation



4096-QAM Modulation

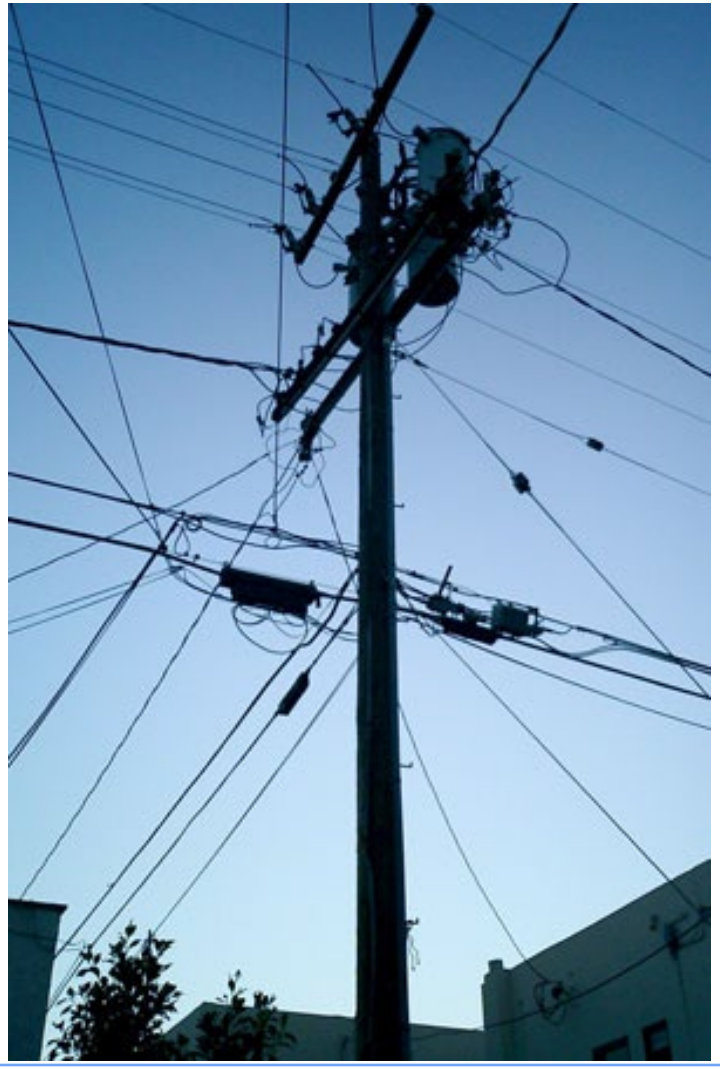


FTTx: Fiber to the curb/home/node

- A reasonable solution for new properties
  - optical fiber is cheap, termination costs dropping
- Digging up the street is very expensive
  - Especially digging into every home
- Fiber to the 'curb-side box'
  - remaining short length of existing copper good for 100's of Mbps.

What is FTTx?

- Mostly a telco issue
- Telcos have many ageing cable plants
- CATV companies have HFC networks in place
- Competitive carriers now install own fiber
- Telcos fighting for exclusive rights

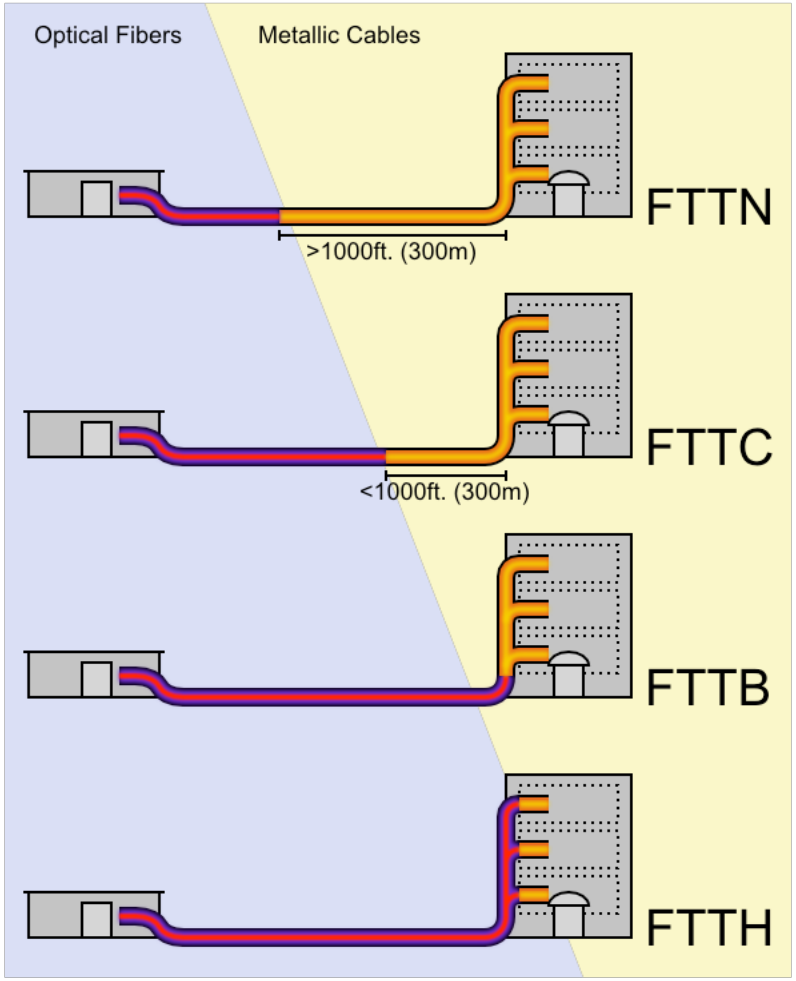


What is FTTx?



- FTTH: fiber to the home
- FTTP: fiber to the premises
- FTTN: Fiber to the node
- FTTC: fiber to the curb
- FTTB: Fiber to the building
- FTTR: Fiber to the Room
- FTTx: for those who can’t decide what to call it or are referring to all varieties!

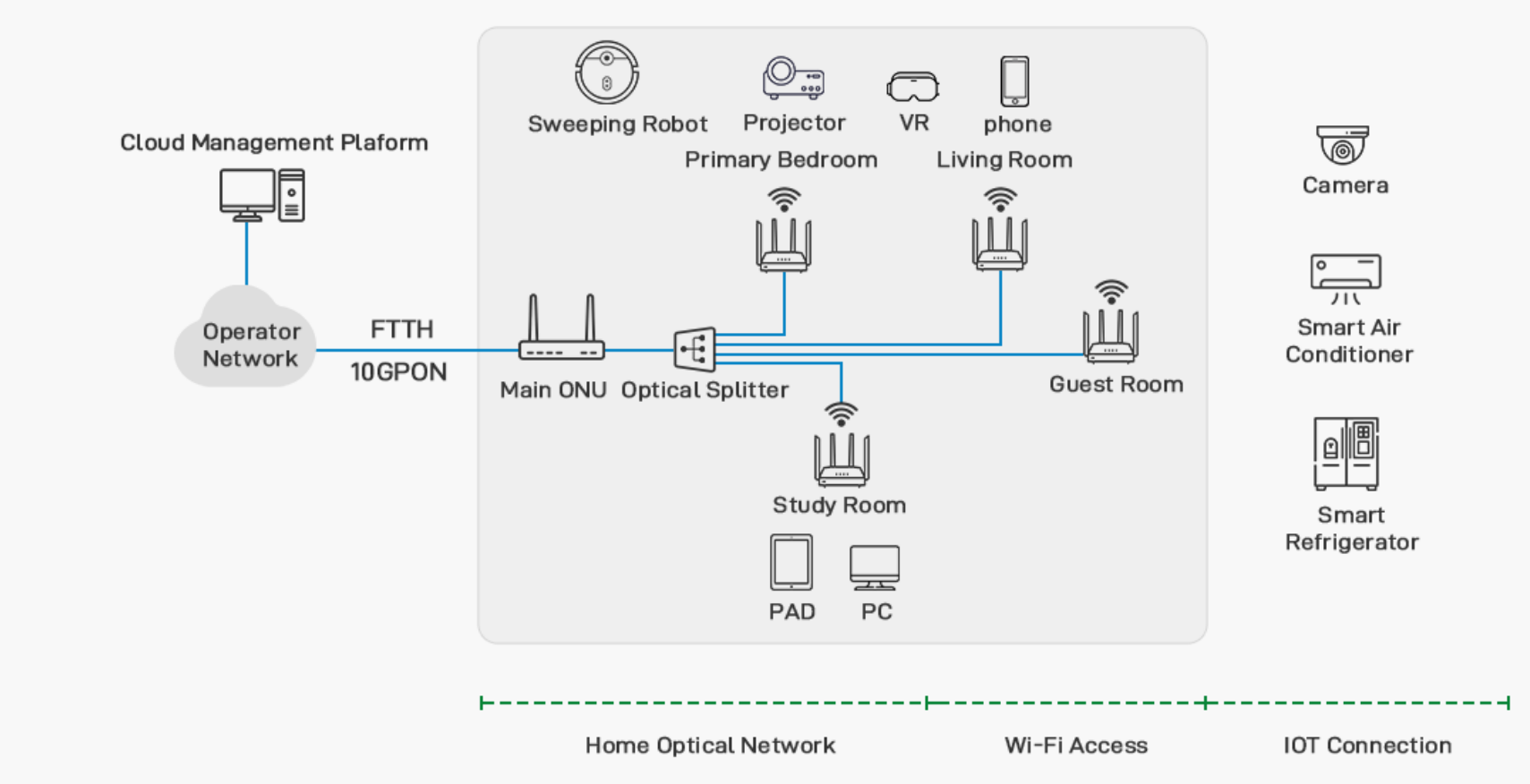
FTTx – Typical Deploys



- FTTN: Fiber to the node
- FTTC: fiber to the curb
- FTTB: fiber to the building
- FTTH: fiber to the home

Fiber to the Room (FTTR) Deployment

- Fiber to the Room (FTTR with FTTH)



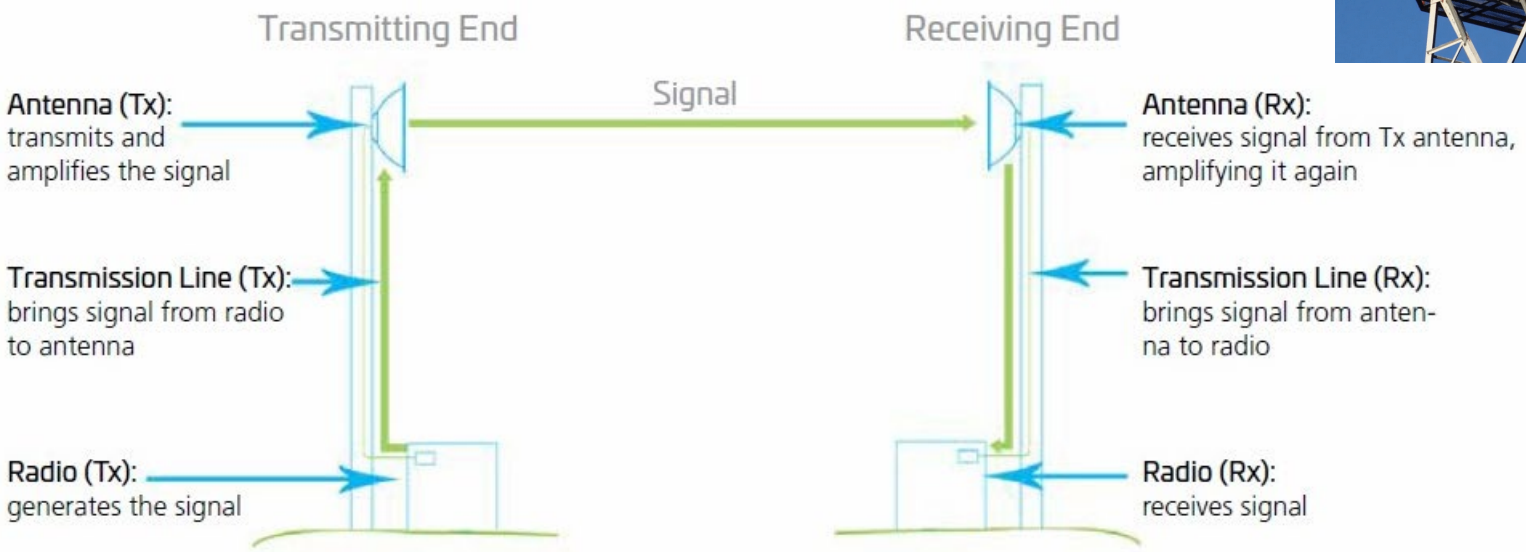
Fixed Wireless

- Microwave (1 GHz to 300 GHz) and free-space laser
  - line-of-sight between antennas
    - avoids multi-path interference, lower power
- Free-space laser systems
  - 2-155Mbps and up
  - relatively narrow beam requires stable fixtures
  - Wavelength Division Multiplex systems
- Some applications
  - Satellite Communication: Communication with satellites in orbit, used for TV, radio, and internet services.
  - Cellular Networks: Backhaul communication between base stations.
  - Point-to-Point Links: Communication between two fixed locations, commonly used in business, government, and military networks.
  - Radar Systems: Used in weather forecasting, air traffic control, and military applications.
- Challenges:
  - Requires clear line of sight.
  - Suitable to weather conditions, such as rain or fog, leading to signal attenuation.
  - Equipment can be costly, especially for high-frequency systems.

Fixed Wireless

- Basic diagram





Microwave Communication Basics eBook

Mobile Wireless - Public Mobile

- Licensed Spectrum provided by government
- Constant Evolution since 1980
  - 1G – 1980; 2G – 1990; 3G – 2000; 4G -2010; 5G – 2020
- 2G – Different standards
  - D-AMPS, GSM and CDMA
  - Global System for Mobile Communications (GSM)
    - originally a digital, circuit-switched network optimized for full duplex voice telephony
    - Expanded to include data communication
      - General Packet Radio Service (GPRS)
      - EDGE(Enhanced Data GSM Evolution)
    - GSM provided 9600 and 14400bps circuit data service
      - Slow connection setup, no stat-mux gain, 600ms RTT
    - GPRS – packet data over GSM
      - 32Kb/s - 100Kb/s, 900-1500ms RTT!
- 3G – Universal Mobile Telecommunications System (UMTS)
  - Also known as International Mobile Telecommunications-2000 (IMT-2000)
  - High-Speed Downlink Packet Access (HSDPA) and High-Speed Uplink Packet Access (HSUPA)
  - HSPA+(Evolved High-Speed Packet Access)
- 4G – Long-Term Evolution
  - International Mobile Telecommunications-Advanced (ITU-T)
- 5G – Fifth Generation of Mobile Telephony
  - International Mobile Telecommunications-2020 (ITU-T)

Mobile Wireless Evolution

- From 2G to 4G

Technology	Full Name	Max Data Speed (Downlink)	Max Data Speed (Uplink)	Release Year	3GPP Standard	Key Features
2G (GSM)	Second Generation, GSM	Up to 64 Kbps	N/A	1991	2G (GSM)	Basic voice and SMS services, limited data capabilities (e.g., SMS, MMS).
2.5G (GPRS)	General Packet Radio Service	Up to 115 Kbps	N/A	2000	2.5G (GPRS)	Packet-switched data, enabling simple internet browsing, email, and MMS.
2.75G (EDGE)	Enhanced Data Rates for GSM Evolution	Up to 384 Kbps	N/A	2003	2.75G (EDGE)	Enhanced GPRS for better data throughput, supporting basic web services.
3G (UMTS/WCDMA)	Universal Mobile Telecommunications System	Up to 2 Mbps	384 Kbps	2001	3G (WCDMA/UMTS)	Enabled mobile internet, video calling, and multimedia content at faster speeds.
H (HSPA)	High-Speed Packet Access	Up to 14.4 Mbps	5.8 Mbps	2005	3G (HSPA)	Higher speeds for better mobile browsing, media streaming, and video calls.
H+ (HSPA+)	Evolved High-Speed Packet Access	Up to 42 Mbps	11 Mbps	2007	3G (HSPA+)	Further improved data speeds, optimized for better mobile services.
H+ Release 6	HSPA+ Release 6	Up to 42 Mbps	N/A	2008	3G (HSPA+ Release 6)	Initial HSPA+ release, improving data rates and user experience.
H+ Release 7	HSPA+ Release 7	Up to 84 Mbps	N/A	2010	3G (HSPA+ Release 7)	Enhanced data throughput, optimized network capacity and efficiency.
H+ Release 8	HSPA+ Release 8	Up to 168 Mbps	N/A	2011	3G (HSPA+ Release 8)	Increased speeds and improved network performance, approaching 4G speeds.
H+ Release 9	HSPA+ Release 9	Up to 168 Mbps	42 Mbps	2012	3G (HSPA+ Release 9)	Further improvements in capacity and user experience in dense areas.
H+ Release 10	HSPA+ Release 10	Up to 168 Mbps	42 Mbps	2013	3G (HSPA+ Release 10)	Final major HSPA+ release, with optimized spectrum use and better throughput.
4G LTE	Long-Term Evolution	Up to 100 Mbps	50 Mbps	2009	4G (LTE)	True 4G technology, offering high-speed mobile broadband with low latency.
4G LTE-A	LTE-Advanced	Up to 1 Gbps	150 Mbps	2013	4G (LTE-A)	Enhanced LTE with carrier aggregation for higher speeds, lower latency.

Mobile Wireless - Evolution

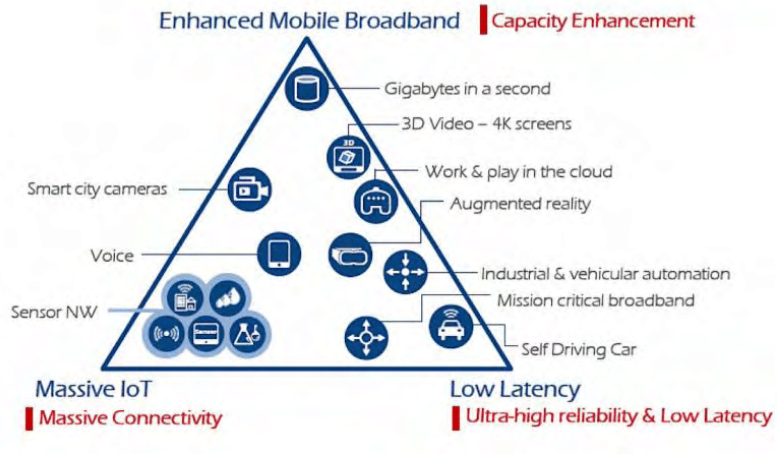
- Another Comparison
  - Different applications Enabled

Comparison	2G	3G	4G	5G
Introduced in year	1993	2001	2009	2018
Technology	GSM	WCDMA	LTE, WiMAX	MIMO, mm Waves
Access system	TDMA, CDMA	CDMA	CDMA	OFDM, BDMA
Switching type	Circuit switching for voice and packet switching for data	Packet switching except for air interference	Packet switching	Packet switching
Internet service	Narrowband	Broadband	Ultra broadband	Wireless World Wide Web
Bandwidth	25 MHz	25 MHz	100 MHz	30 GHz to 300 GHz
Advantage	Multimedia features (SMS, MMS), internet access and SIM introduced	High security, international roaming	Speed, high speed handoffs, global mobility	Extremely high speeds, low latency
Applications	Voice calls, short messages	Video conferencing, mobile TV, GPS	High speed applications, mobile TV, wearable devices	High resolution video streaming, remote control of vehicles, robots, and medical procedures

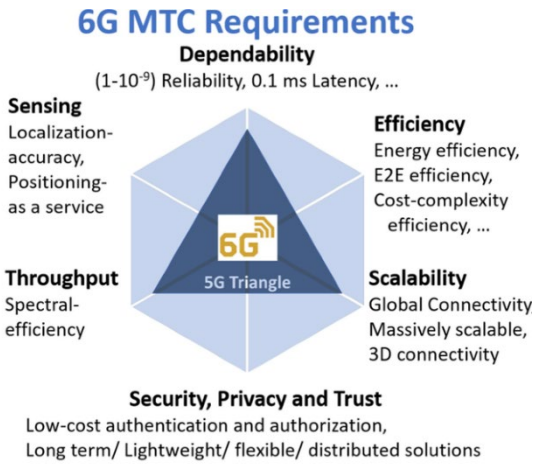
Comparison of 2G 3G 4G 5G | 2G vs 3G vs 4G vs 5G | 5G vs 4G | Difference between 4G and 5G | Rantcell

Mobile Wireless
Some Evolution Trends

- Application and services must be capable of satisfying its user's requirements
  - This is also true for the network and the infrastructure
- Networks are evolving
  - SDN (2009), Workspace (2010); NFV (2012); P4 (2014); Intents (2016);
  - IMT Vision (2016) - eMMB, URLLC, mMTC (5G Triangle);
  - 6G-Vison (2021) – 5G+Security, Privacy, Trust; Efficiency; Sensing (6G Hexagon)
- The evolution happens toward to fulfill specific applications requirements
- The 5G to 6G evolution trend



The METIS 5G System Concept: Meeting the 5G Requirements, 2016
DOI: 10.1109/MCOM.2016.1500799CM



Machine type communications: key drivers and enablers towards the 6G era, 2021
DOI: 10.1186/s13638-021-02010-5

Mobile Wireless Evolution of Standards

- Evolution
- Current evolution happens inside 3GPP
- 5G
  - 3GPP
    - Release 16
    - Release 17
    - Release 18
      - 5G Advanced
  - The 3GPP's System of Parallel Releases
  - 3GPP Releases History

The evolution of mobile standards

Mobile standards	3GPP		Qualcomm	China	IEEE
Carriers using:	AT&T and T-Mobile US, majority of global carriers		Sprint, Verizon Wireless	China Mobile	Sprint
2G: digital + data services	GSM: 2G		CDMAOne		
	GPRS: 2.5G				
	EDGE: 2.75G				
3G: at least 200 kbps iPhone 4 currently delivers up to 7.2Mbps down, 5.8Mbps up	Release 4	UMTS 3G	CDMA2000 EVDO rev 0	TD-SCDMA (up to 2Mbps)	
	Release 5	HSDPA 3.5G (to 21Mbps down)	CDMA2000 EVDO rev A (up to 3.1Mbps down, 1.8 up)		
	Release 6	HSUPA 3.5G (to 5.8Mbps up)	EVDO Rev C / Ultra Mobile Broadband Canceled: Sprint moving to WiMAX, Verizon moving to 3GPP LTE		Mobile WiMAX 3.9G (4 Mbps cap on EVO "4G")
	Release 7	HSPA+ 3.5G			
	Release 8/9	LTE 3.9G			
4G: at least 100 Mbps, IP-based	Release 10	LTE Advanced		TD-LTE	WiMAX 4G

NPD ranks iPhone 4S as America's most popular 4G phone due to HSPA+ | AppleInsider, 2012

Mobile Wireless

- Evolution

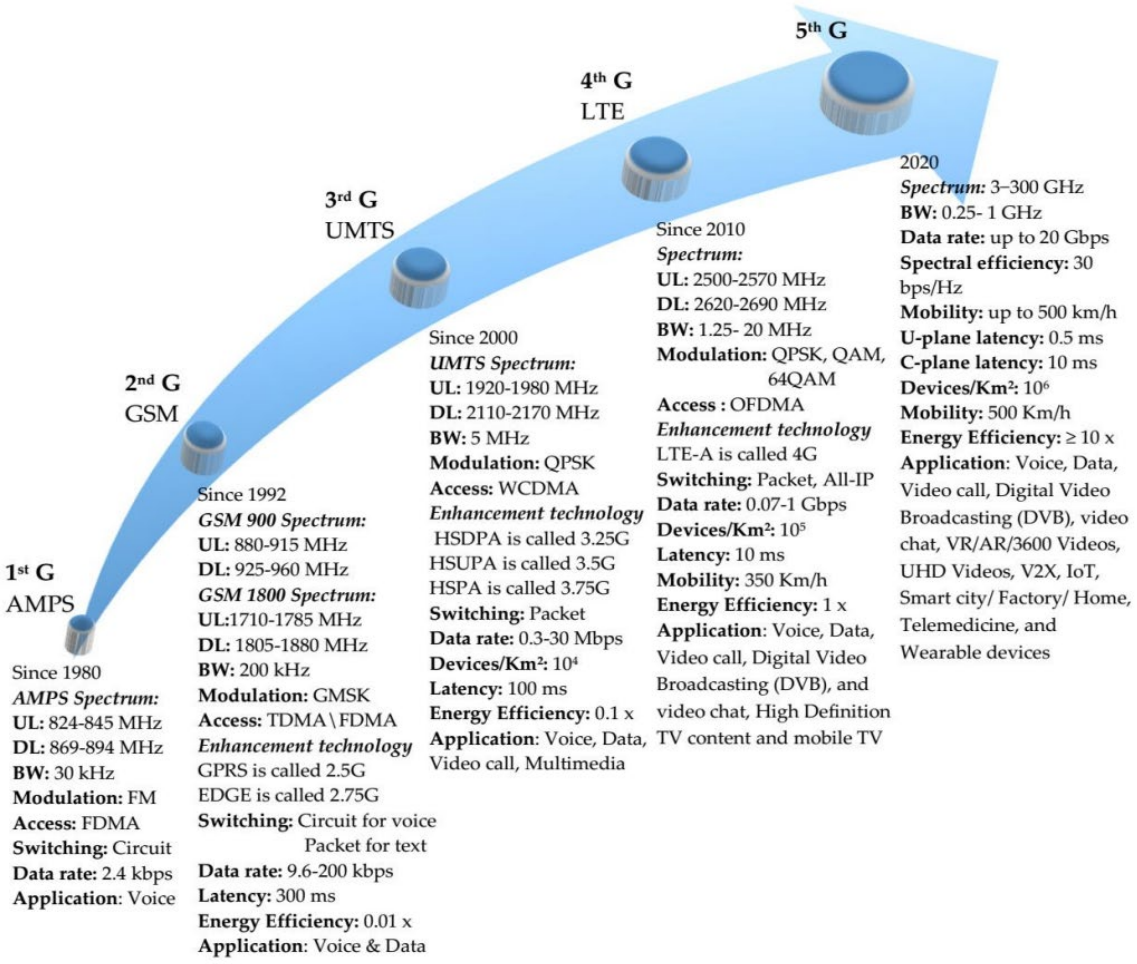


Figure 1. Leading communication achievements for several generations (1G to 5G)

[Evolution of wireless communication networks: from 1G to 6G and future perspective, 2022]

IEEE 802.11x – Wi-Fi

- Family of wireless network protocols based on the IEEE 802.11 family of standards
- Local area networking
  - Home and Enterprises
  - Small areas
- Mobility problem

Wi-FI Evolution

- Lastest release is Wi-Fi 7
- Constant evolution
- Wi-Fi 8 planned for 2028
  - Focus on reliability
  - new possibilities for advanced applications.
  - With its potential for lower latency and more stable connections
  - virtual and augmented reality, real-time gaming, and industrial automation

Wi-Fi generations V·T·E

Generation	IEEE standard	Adopted	Maximum link rate (Mbit/s)	Radio frequency (GHz)
Wi-Fi 8[44]	802.11bn	2028	100,000[45]	2.4, 5, 6, mmWave
Wi-Fi 7	802.11be	2024	1376–46,120	2.4, 5, 6
Wi-Fi 6E	802.11ax	2020	574–9608[46]	6[b]
Wi-Fi 6		2019		2.4, 5
Wi-Fi 5	802.11ac	2014	433–6933	5[c]
Wi-Fi 4	802.11n	2008	72–600	2.4, 5
*(Wi-Fi 3)**	802.11g	2003	6–54	2.4
*(Wi-Fi 2)**	802.11a	1999		5
*(Wi-Fi 1)**	802.11b	1999	1–11	2.4
*(Wi-Fi 0)**	802.11	1997	1–2	2.4

**Wi-Fi 0, 1, 2, and 3 are named by retroactive inference.*

Wi-Fi - Wikipedia

Wi-Fi Evolution

- Generations

Wi-Fi Standard	Year Introduced	Frequency Bands	Max Speed (Theoretical)	Range	Modulation	Key Features
Wi-Fi 1 (802.11b)	1999	2.4 GHz	11 Mbps	Short (about 30 m)	DSSS (Direct Sequence Spread Spectrum)	First widely adopted standard; limited speed and range.
Wi-Fi 2 (802.11a)	1999	5 GHz	54 Mbps	Short (about 35 m)	OFDM (Orthogonal Frequency Division Multiplexing)	First to support 5 GHz, less prone to interference, but shorter range than 802.11b.
Wi-Fi 3 (802.11g)	2003	2.4 GHz	54 Mbps	Medium (about 45 m)	OFDM	Combines the best of 802.11a (speed) and 802.11b (range). Compatible with older standards.
Wi-Fi 4 (802.11n)	2009	2.4 GHz, 5 GHz	600 Mbps	Long (about 70 m)	MIMO (Multiple Input Multiple Output), OFDM	Introduced MIMO technology, better speed, range, and reliability, dual-band support.
Wi-Fi 5 (802.11ac)	2014	5 GHz	3.5 Gbps	Long (about 100 m)	MU-MIMO (Multi-User MIMO), 256-QAM	Focused on high-speed internet, uses wider channels (80 MHz/160 MHz), and introduced MU-MIMO for better efficiency.
Wi-Fi 6 (802.11ax)	2019	2.4 GHz, 5 GHz	9.6 Gbps	Long (about 100 m)	OFDMA (Orthogonal Frequency Division Multiple Access), 1024-QAM	Enhanced performance in crowded areas, increased capacity, better efficiency with many devices, improved security.
Wi-Fi 6E (802.11ax)	2021	6 GHz	9.6 Gbps	Long (about 100 m)	OFDMA, 1024-QAM	Expansion of Wi-Fi 6 into the 6 GHz band, offering more channels, less interference.
Wi-Fi 7 (802.11be)	2024	2.4 GHz, 5 GHz, 6 GHz	46 Gbps	Long (about 100 m)	4096-QAM, Multi-AP Coordination	Up to 16 spatial streams, faster speeds, lower latency, better for applications like AR/VR, and ultra-high-definition video.

Satellite

- GEO (Geostationary Earth Orbit) stationary
  - 36,000km orbit
  - long delays, e.g. 2x 120ms RTT
- MEO (Medium Earth Orbit)
  - Altitude 5,000 to 20,000km
  - historically been used for GPS and other navigation applications
- LEO (Low Earth Orbit) constellations
  - 20+ in 1,500km orbits (2hr)
  - latency typically sub 100ms, 300Mbps+
  - interconnect options:
    - Forward to ground station
    - Uplink to a GEO network
    - LEO to LEO laser

GEO, MEO, LEO Comparison

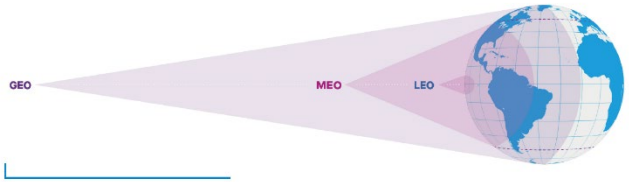


Figure 1: Schematic of orbital altitudes and coverage areas

	GEO (36,000km)	MEO (5,000-20,000km)	LEO (500-1,200km)
Altitude latency[1]	High	Low	Very low
Earth coverage	Very large	Large	Small
Satellites required	Three	Six	Hundreds
Data gateways	Few fixed	Regional flexible	Local numerous
Antenna speed	Stationary	1-hour slow tracking	10-minute fast tracking
Advantages	High throughput (HTS) technologies enable basic broadband internet applications	Proven low latency comparable to terrestrial networks, offers fibre-equivalent performance	Claims support for high-frequency trading, virtual gaming, and high-performance computing applications
	Fewer satellites over very large fixed geographical areas	Simple equatorial orbit covers 96% of global population	Smaller, lower power satellites batch-launched more cheaply than GEO
Disadvantages	High altitude and distant ground networking impacts latency-sensitive applications	Dual tracking antennas required to maintain continuous connectivity	Very complex tracking and ground network, plus complete constellation must be in place before service starts
	Signal power losses require larger satellites and antennas	Inclined plane orbits needed to cover high latitudes	Unproven business model, risky technology, and space debris risk

[1]*Total end-to-end network latency is dependent on ground infrastructure*

Table 1: Comparison of GEO, MEO, and LEO satellites

[GEO, MEO, and LEO - Via Satellite (satellitetoday.com)]